



TARGET-90

JKBOSE CLASS-11TH

Morphology of Flowering plants

One Shot

Botany

ARSHID Sir



Topics To Be Covered

1 One Shot

2 ...

Introduction

Morphology — Study of external features of an organism.

Angiosperms (flowering plants)

- a) Root
 - b) Stem
 - c) Leaf
 - d) Flower
 - e) Fruit
 - f) Seed
- vegetative parts (veg. Morphology).
- Reproductive parts (Reproductive morphology).

Root

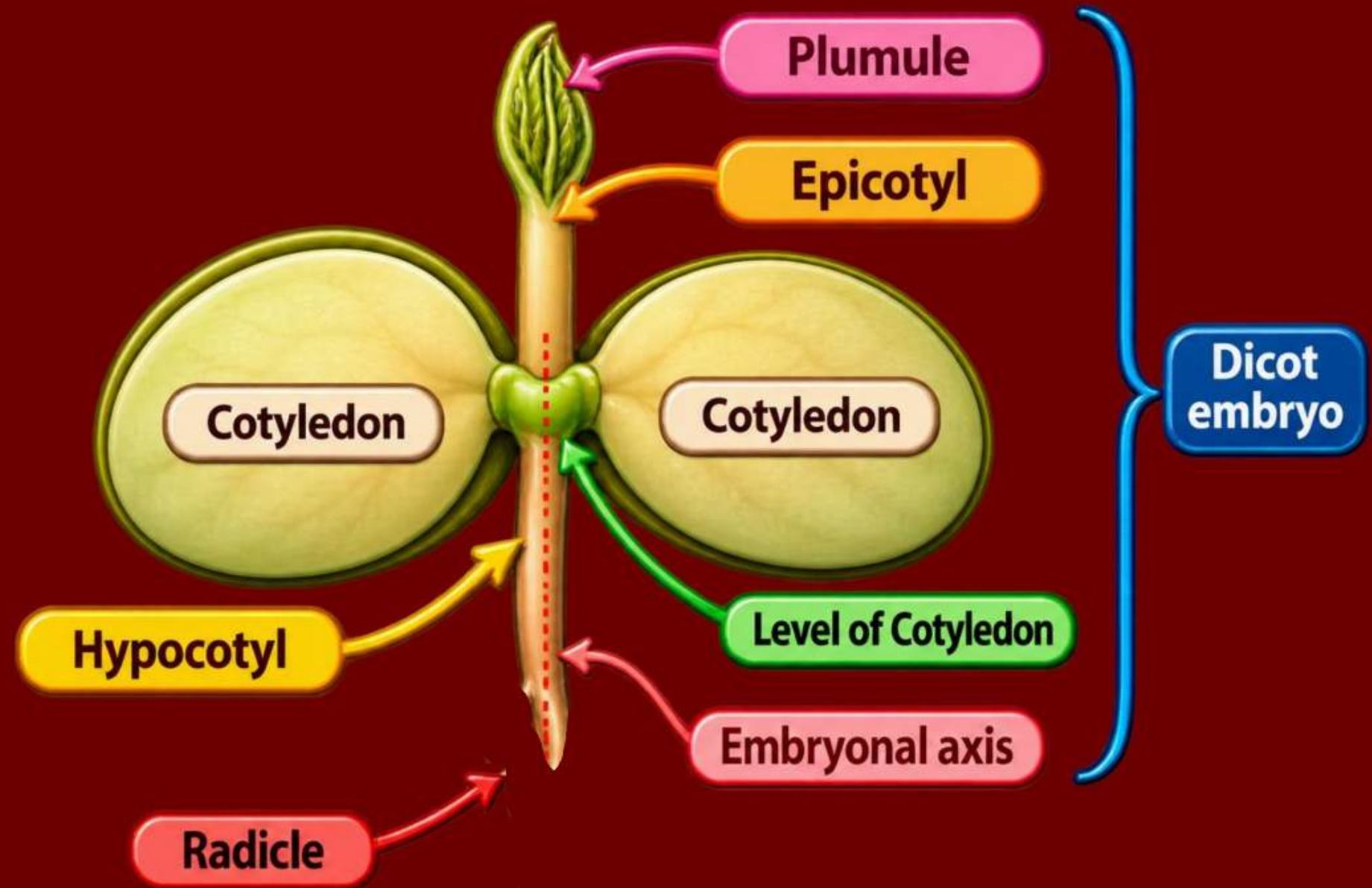
Angiosperms

↓ 02 Divisions

- a. Dicot
- b. Monocot.

Root and Its Functions

a) Dicot ROOT :

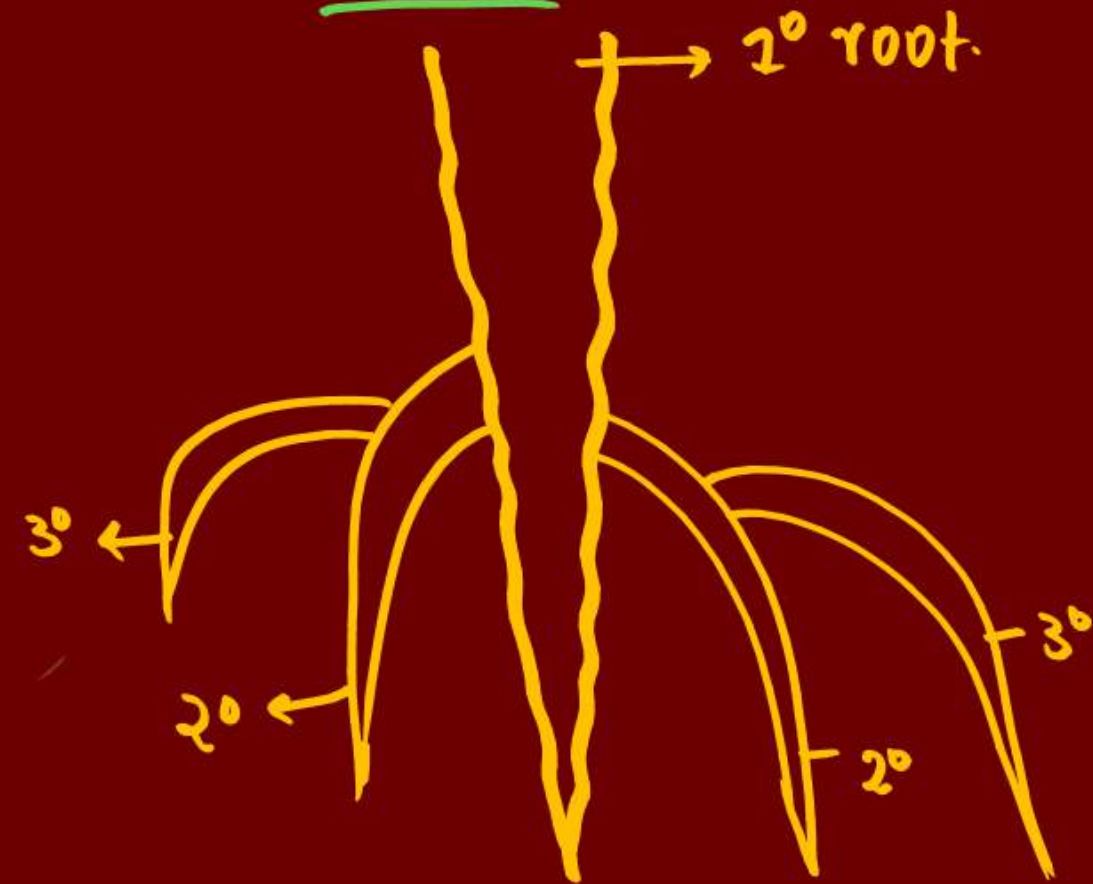


Root and Its Functions

Radicle — direct elongation forms Primary root

- 1° root bears lateral branches of several orders (like 2°, 3°, etc.)
- 1° root along with its lateral branches is called Tap root

Ex. Mustard



Root and Its Functions

b) Monocot Root:

- 1° root formed is short lived
- It is replaced by fibrous roots
- arises from base of stem, numerous
- Example of adventitious root

Ex. Maize, Barley



Root and Its Functions

Adventitious root: root develops from any part of plant other than radicle
in monocots & dicots

Example: Fibrous root, Ficus bengalensis (Banyan tree), Monstera

Functions of Root

1. Provides anchorage to plants.
2. Transports H₂O , minerals from soil.
3. Stores reserve food material.
4. Synthesis of plant growth regulators.

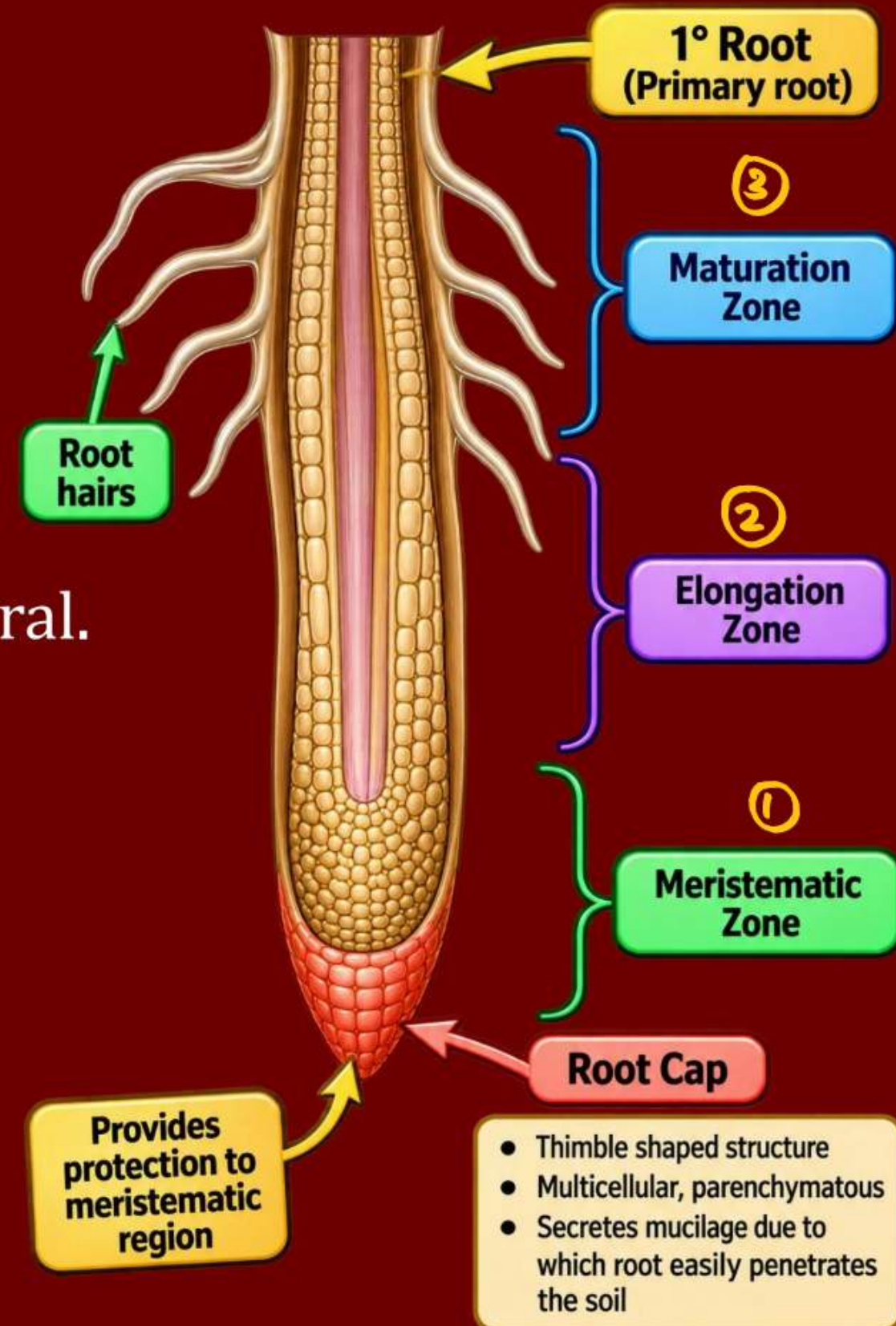
Basic Features of Root

- Root develops from Radicle.
- +vely geotropic.
- +vely hydrotropic.
- -vely phototropic.
- Nodes and internodes - not there.
- Chlorophyll - not there. (No photosynthesis).

Regions of Root

1. Meristematic Zone:

- Cells - dividing.
- Small size, isodiametric.
- Dense cytoplasm - there.
- Vacuole - not there & if there then peripheral.
- Cell wall is primary.
- Multiple plasmodesmatal connections.



Regions of Root



2. Elongation Zone:

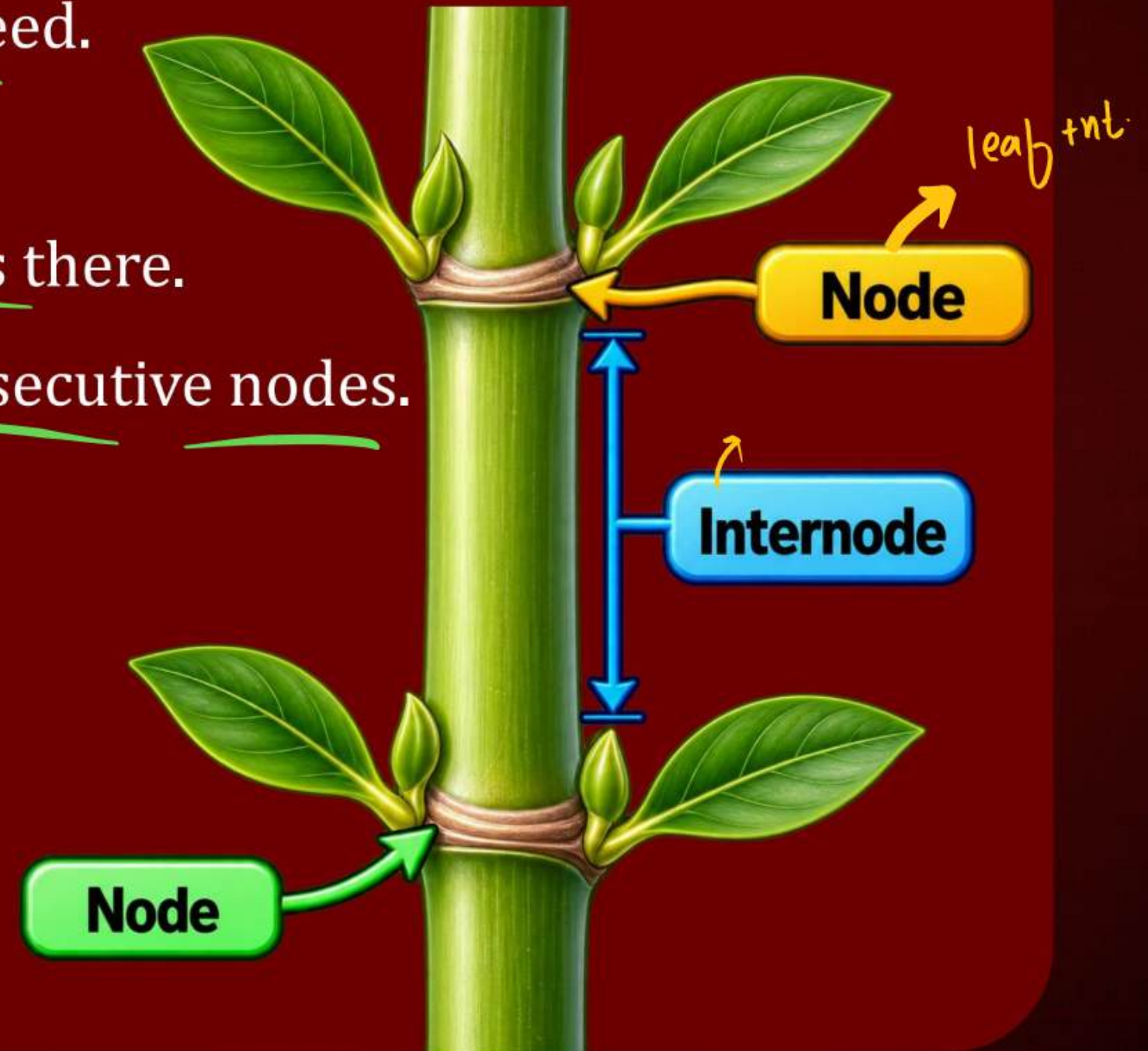
- Here cells elongate.
- It is actually responsible for increase in root length.

3. Maturation Zone:

- Here cells become differentiated & performs particular functions.
- AKA zone of differentiation / Root hair zone.
- Cell wall is thick.
- Cells - permanent.
- Epidermal cells forms root hairs → Increase surface area for absorption of H_2O and minerals.

Stem

- Above ground part of plant.
 - Develops from plumule of germinating seed.
 - Possess Nodes & internodes.
 - **Node:** position on stem at which leaf is there.
 - **Internode:** distance between two consecutive nodes.
- Stem + Branches → Shoot system.



Stem

Stem bears buds:

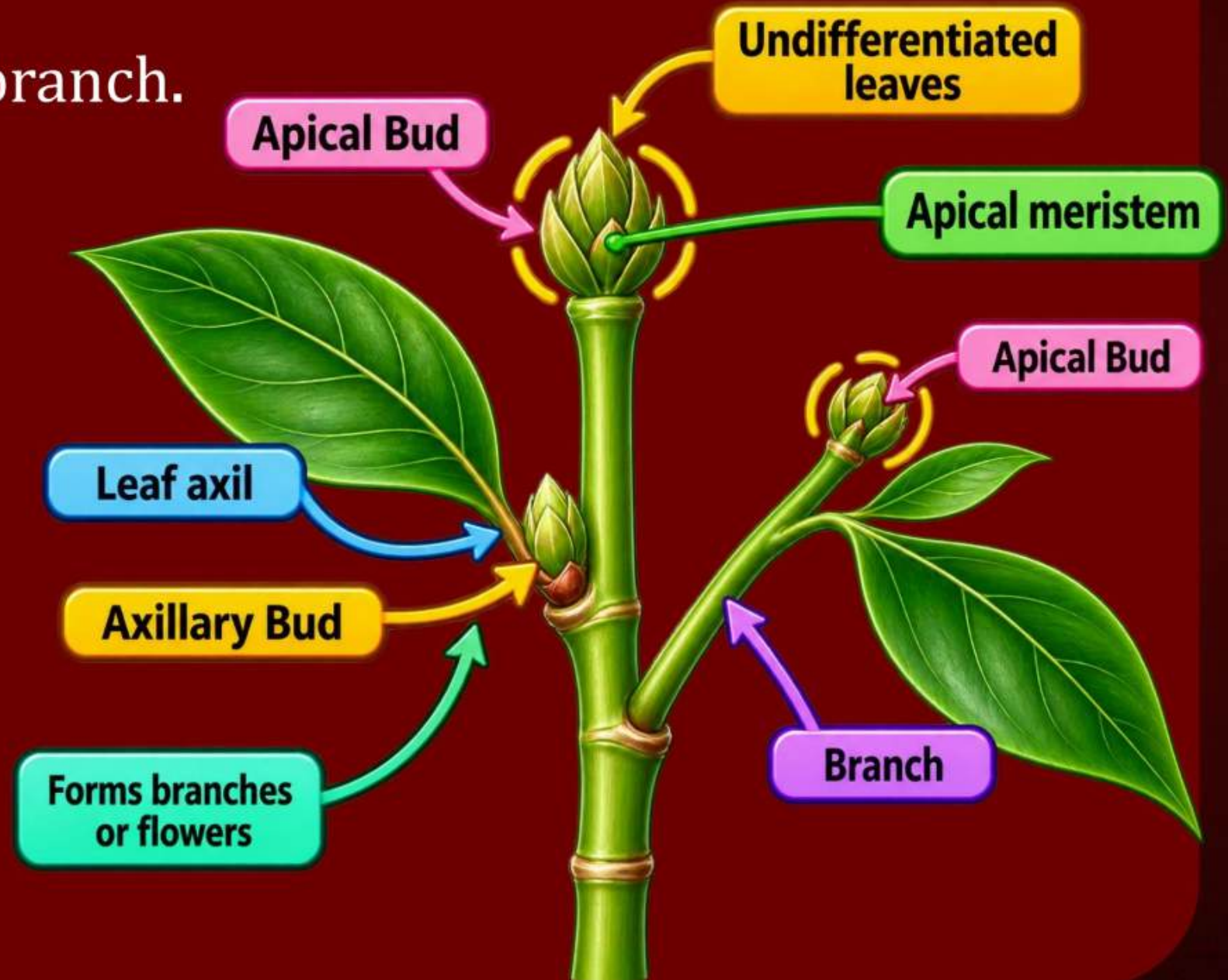
Apical bud or terminal bud

- Present at tip of main stem / tip of branch.

Axillary bud

- Present in axil of leaves.

Forms branches or flowers.



Main Functions of Stem

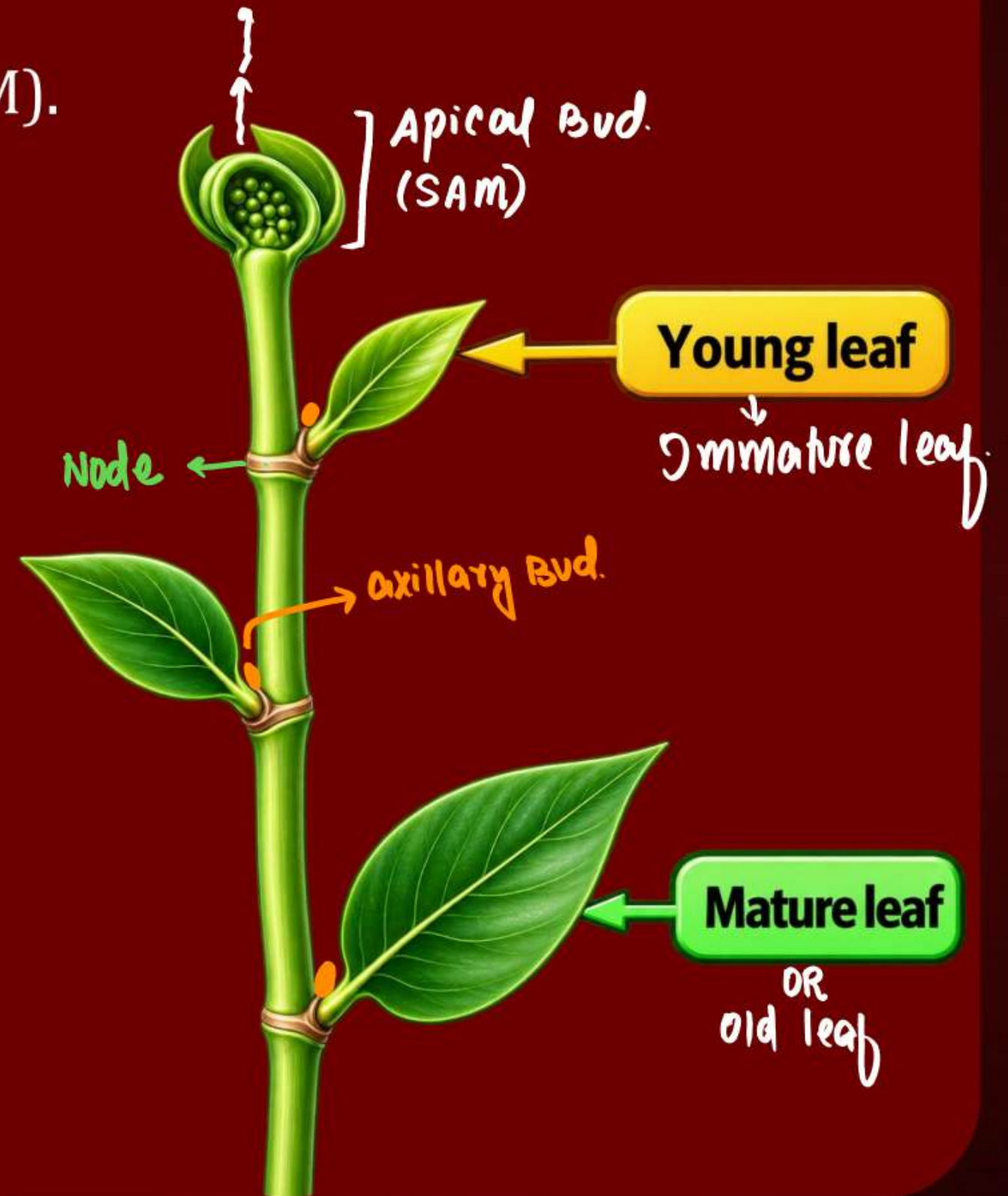
- Stem bears leaves, flowers and fruits.
- Helps in transport of water, minerals and photosynthates (food).
- Some stems store food, gives support and protection.
- Some stems help in vegetative propagation.

Leaf

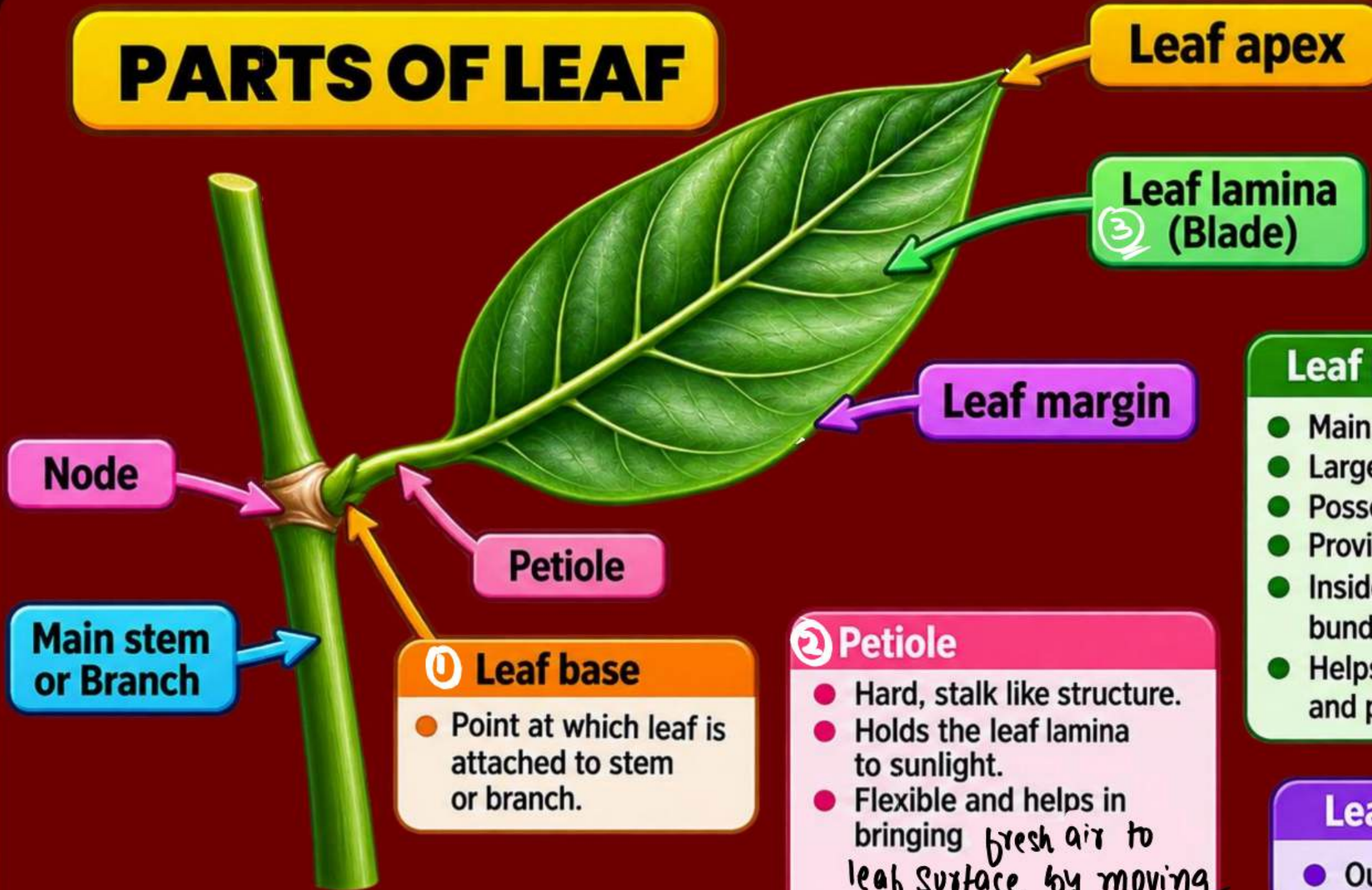
- Develops from shoot apical meristem (SAM).
- Present at nodes of stem.
- Performs photosynthesis.
- Present laterally.
- Arrangement of leaves on stem/branch is

Acropetal

- Young/immature leaf present at apex of stem/branch
- Mature/old leaves present at base of stem/branch



PARTS OF LEAF



Leaf apex

Leaf lamina
(Blade)

Leaf margin

Leaf lamina (Blade)

- Main part of leaf.
- Largest and flatter part.
- Possesses veins or veinlets.
- Provides structural support.
- Inside veins and veinlets are vascular bundles (xylem + phloem).
- Helps in conduction of water, minerals and photosynthate.

Leaf margin

- Outer edge of the leaf lamina.

② Petiole

- Hard, stalk like structure.
- Holds the leaf lamina to sunlight.
- Flexible and helps in bringing fresh air to leaf surface by moving in wind.

① Leaf base

- Point at which leaf is attached to stem or branch.

Node

Main stem
or Branch

Petiole

Venation



Pattern of arrangement of veins & veinlets in leaf lamina.

02 types:

1) Reticulate Venation:

Veins or veinlets form network like structure inside leaf lamina.

Ex. Mostly dicots



Reticulate venation

2) Parallel Venation:

Vein run parallel to each other inside leaf lamina.

Ex. Mostly monocots



Parallel venation

Leaf Types

Leaf Types → 02 types

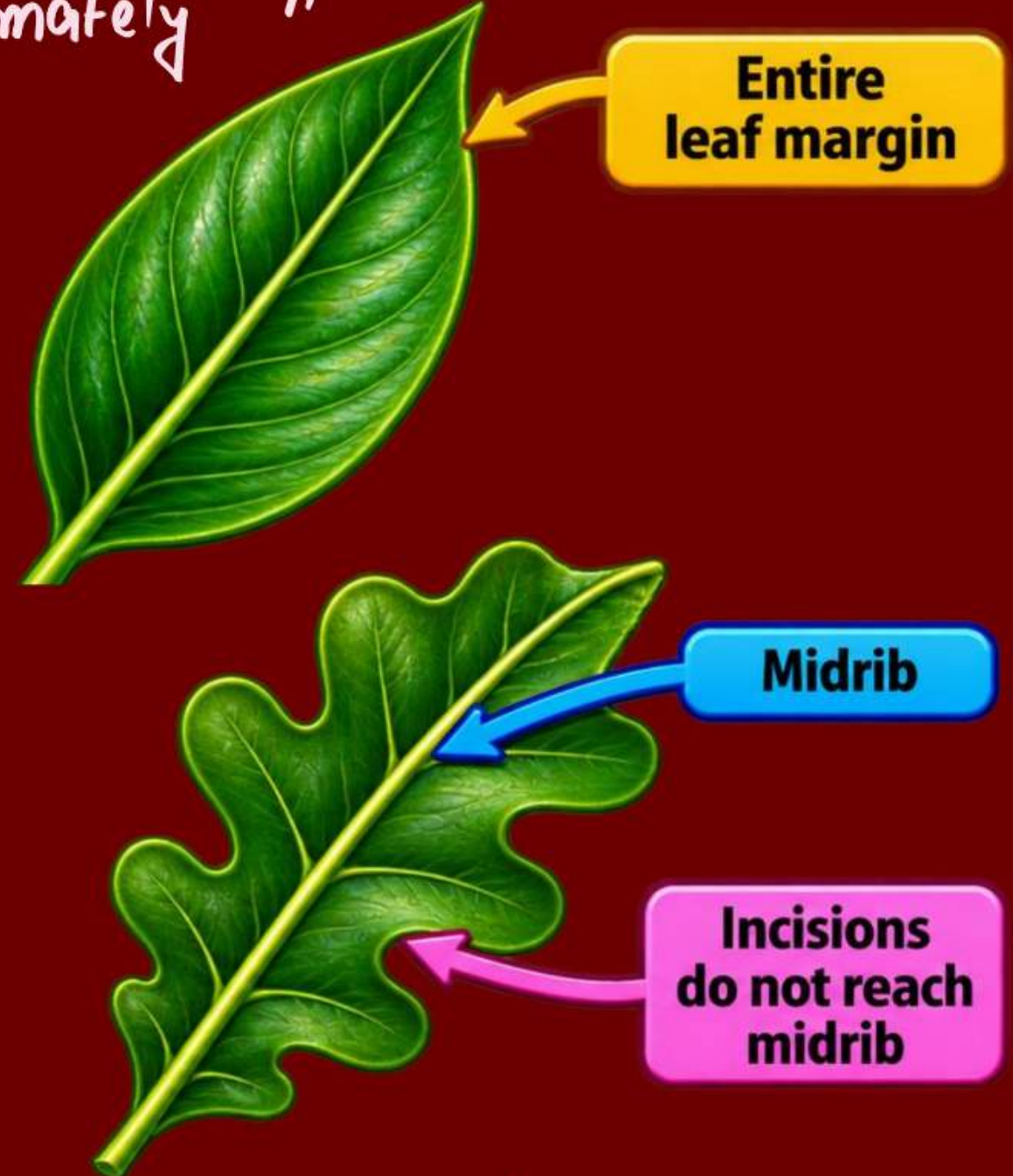
- Simple leaf
- compound leaf
 - Pinnately Compd. leaf
 - palmately "

Simple Leaf:

Leaf margin is entire / incised

Incisions does not reach upto midrib

Ex. Mango.



Leaf Types

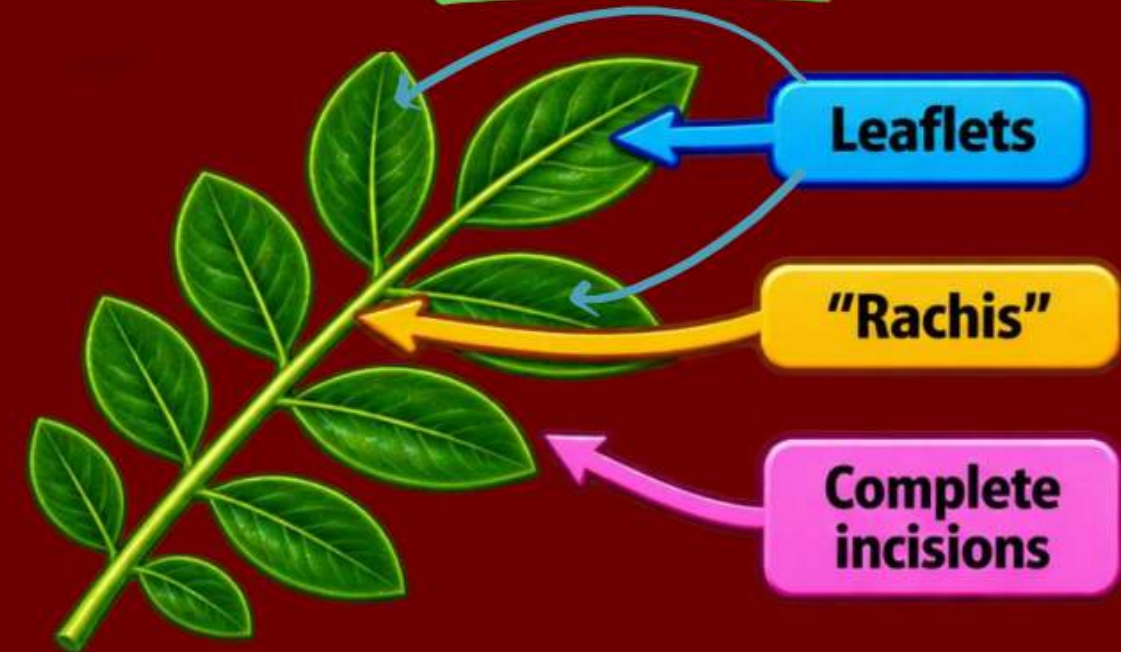
Compound Leaf:

Incisions are present and reaches upto midrib due to which leaf lamina is divided into small leaflets.

Here, midrib now referred as 'Rachis'.

Two types:

- a) Pinnately Compound leaf ✓
- b) Palmately Compound leaf ✓



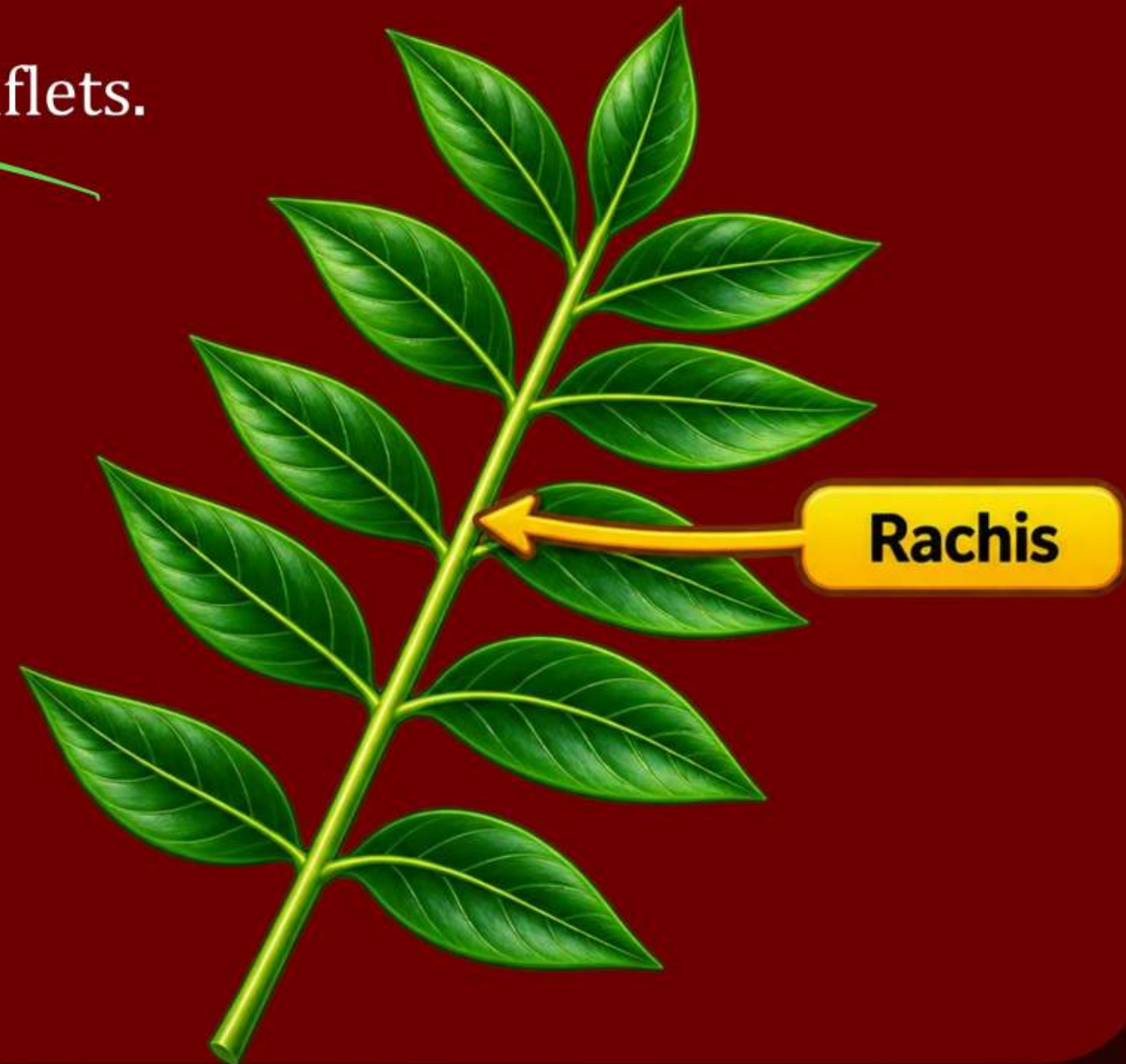
Pinnately Compound Leaf



Pinnately Compound leaf:

Ex. Neem. ✓

Common axis (rachis) bears number of leaflets.



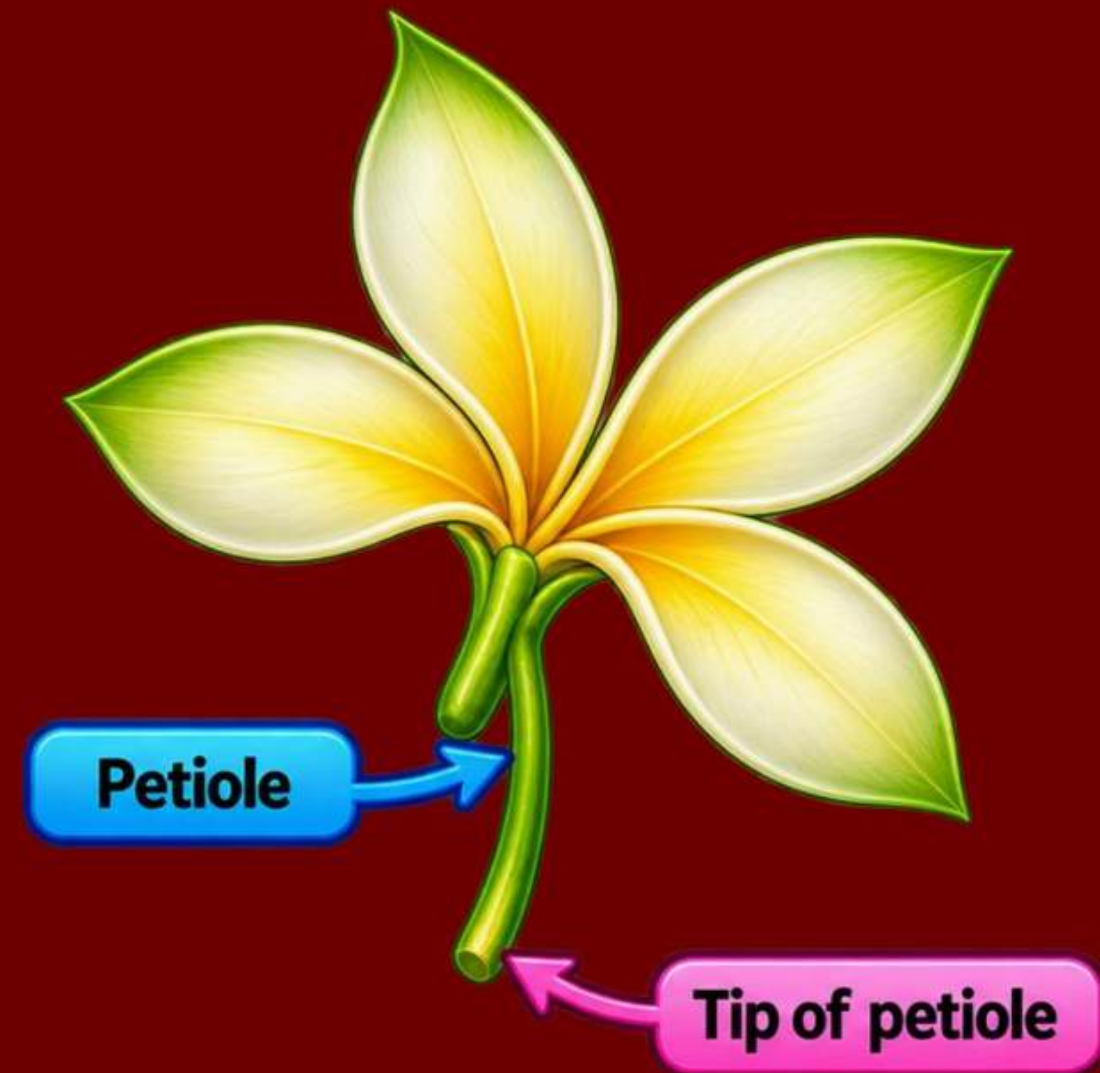
Palmately Compound Leaf

Palmately Compound leaf:

Ex. Bombax (Silk cotton).

Whorl of leaves present at tip of petiole.

Rachis — Absent.



PHYLLOTAXY

Pattern of arrangement of leaves on main stem or branch.

3 types

① ALTERNATE / SPIRAL

China rose, Mustard, Sunflower.

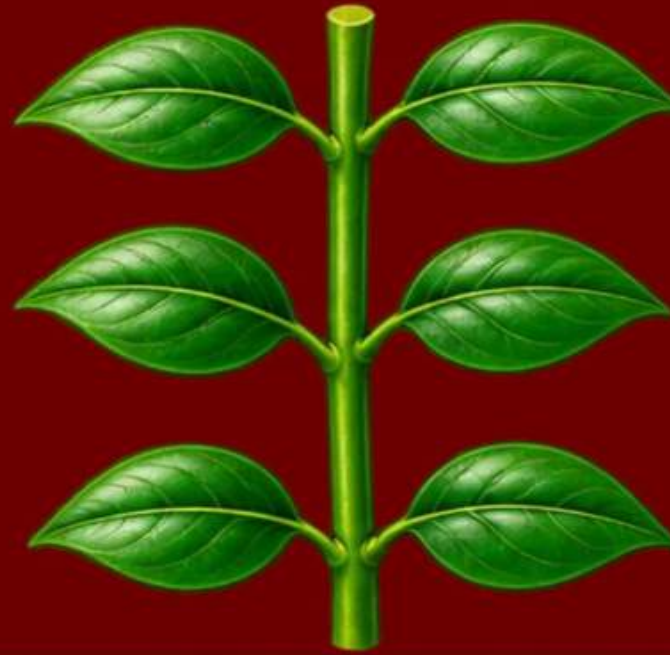


• Most Common.

- Single leaf is present at each node in alternate manner.

② OPPOSITE

Calotropis, Guava.



- At each node two leaves are present which are opposite to each other.

③ WHORLED

Alstonia.



- At each node a whorl of leaves are present.
- Rare.

Inflorescence



Pattern of arrangement of flowers on floral axis or peduncle or inflorescence axis.

Peduncle (may be branch/main stem) or floral axis.

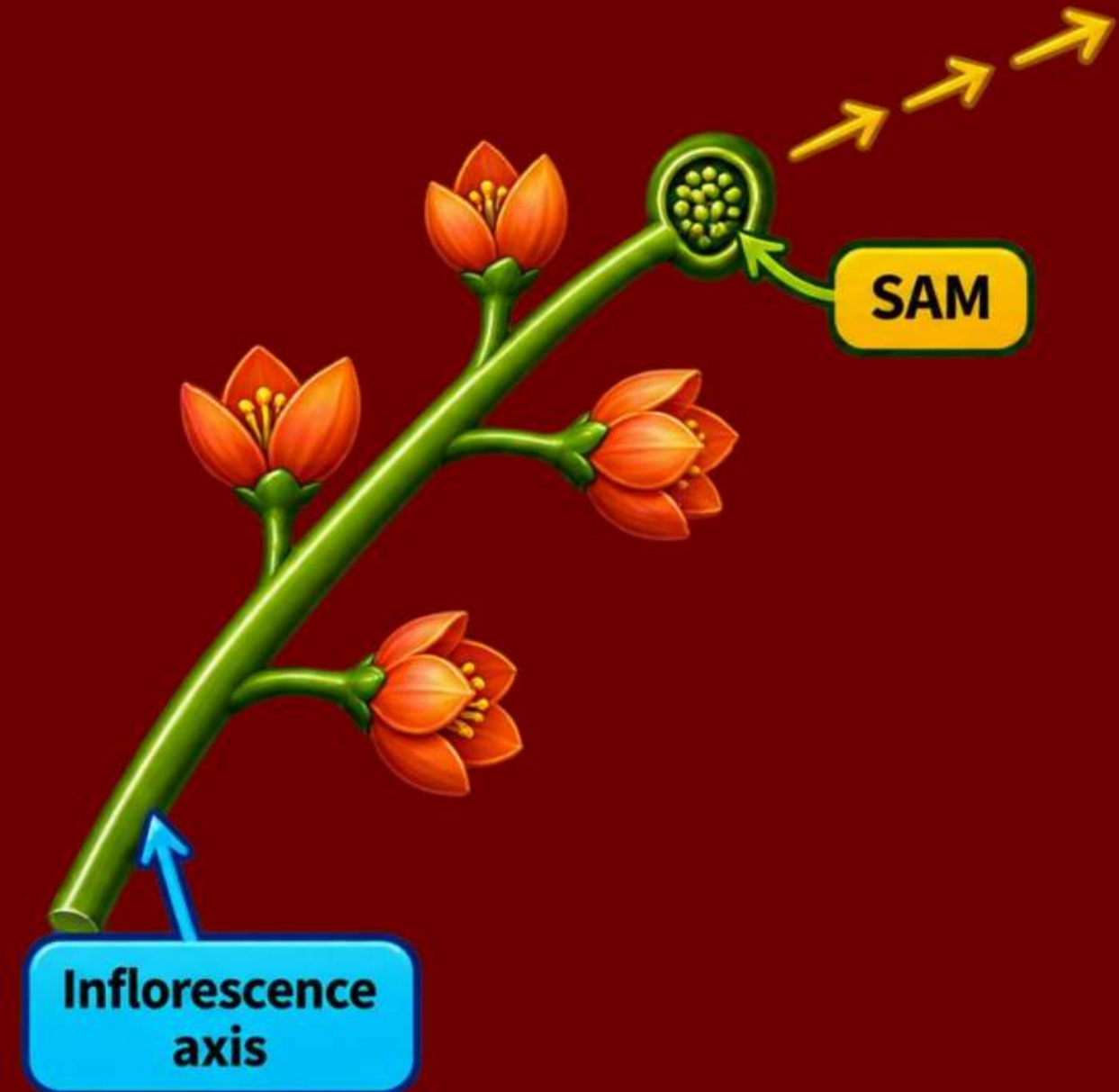
Inflorescence — 03 types:

- 1) Racemose inflorescence
- 2) Cymose inflorescence
- 3) Solitary inflorescence



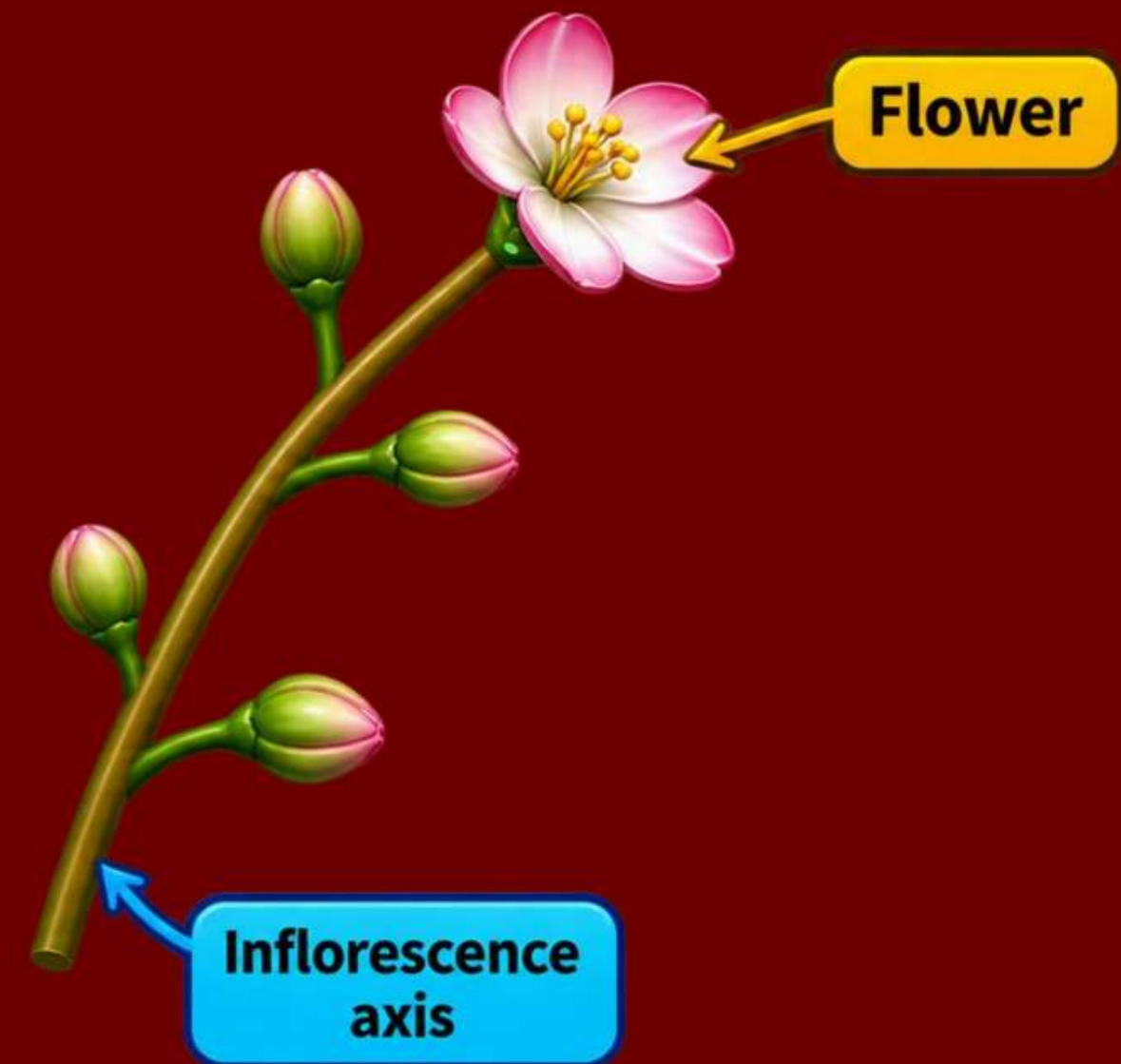
1) Racemose Inflorescence

- At tip of floral axis — SAM is present, due to which it grows continuously → (indeterminate growth) *unlimited growth.*
- Arrangement of flowers is **Acropetal**
 - Young towards apex
 - Old towards base of floral axis
- Ex. Brassicaceae, Fabaceae



2) Cymose Inflorescence

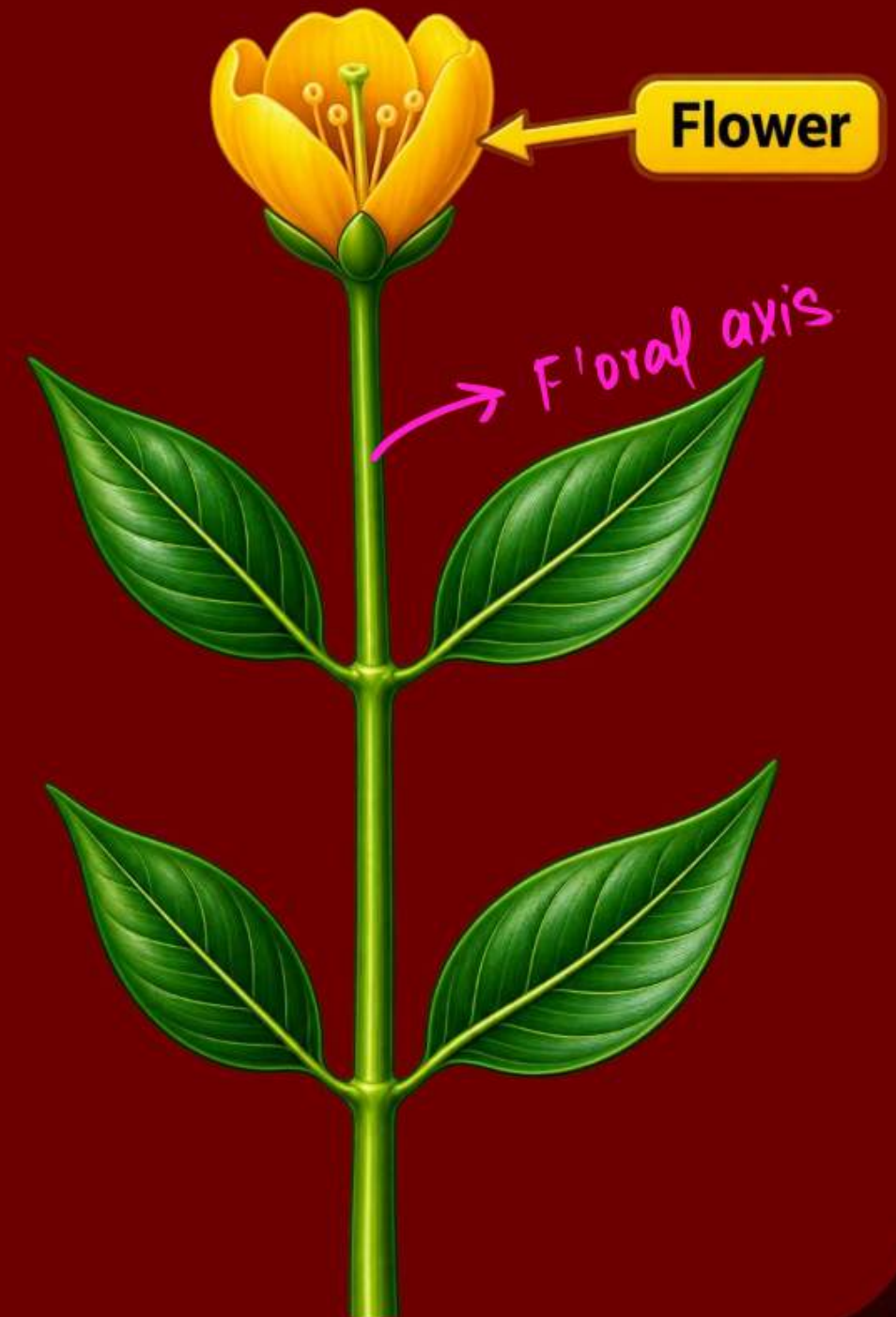
- Floral axis forms flower at its tip due to which its growth is limited (determinate growth)
- Arrangement of flowers is Basipetal
 - Young flowers towards base
 - Mature flowers towards apex
- Ex. Solonaceae ✓
- Ex. Lillaceae ✓



Solitary Inflorescence

presence of single flower at tip of floral axis

Ex. Poppy (Papaver somniform)



Flower — Sexual Reproductive Organ

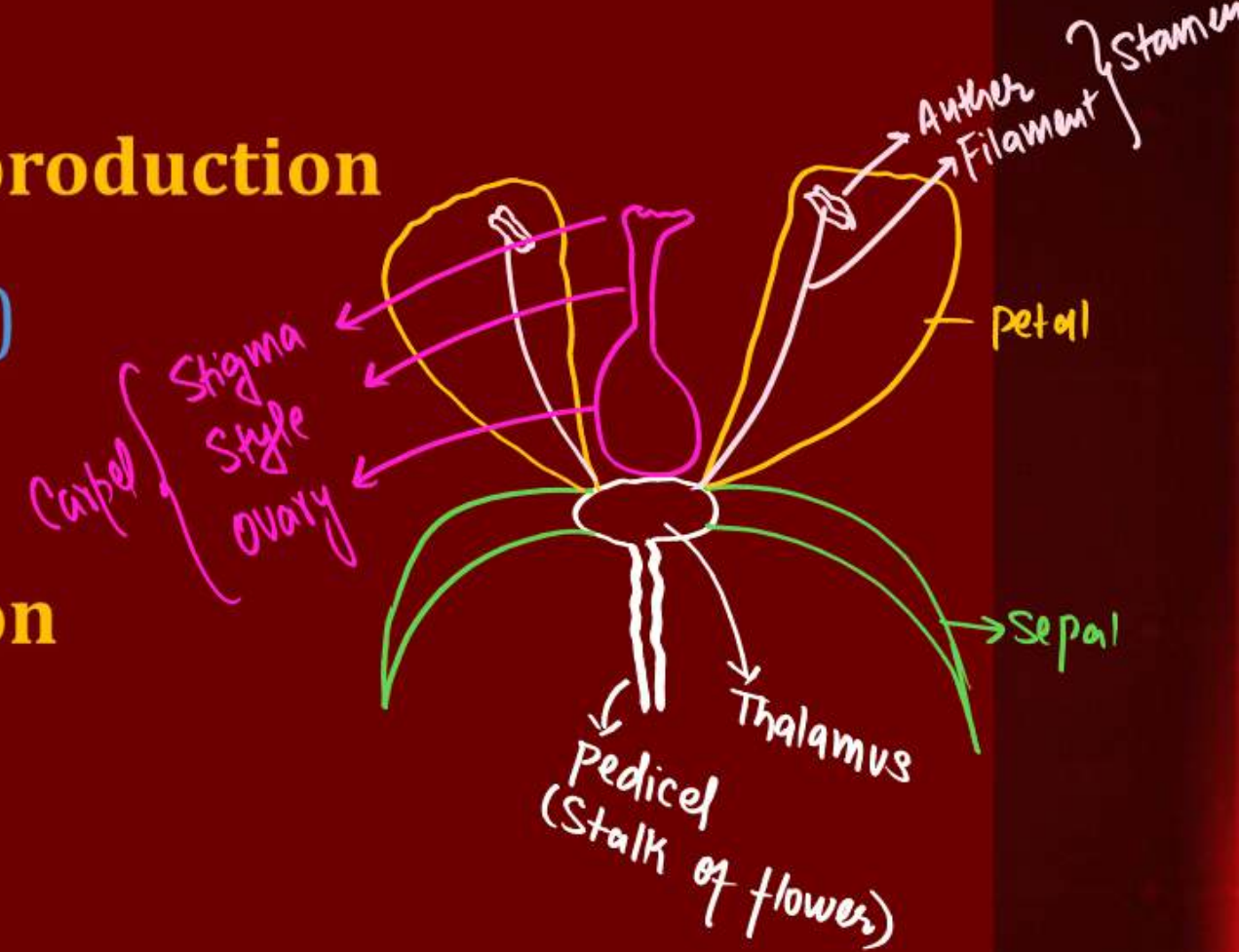
✓ 04 different whorls:

Non-essential whorls :: not involved in sexual reproduction

- a) Calyx (K) — Group of sepals (Outer most whorl)
- b) Corolla (C) — Group of petals

Essential whorls :: involved in sexual reproduction

- c) Androecium (A) — Group of stamens
- d) Gynoecium (G) — Group of Carpels/pistils

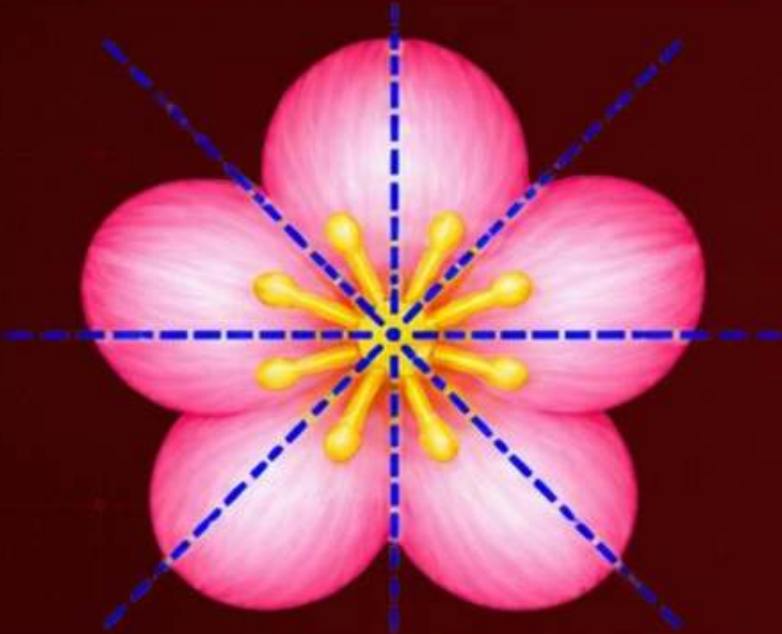


Note: In lilly — Sepals or petals are not distinguishable, so thus referred to as 'TEPALS' Whorl of tepals is referred to as Perianth (P)

FLOWER TYPES (ON THE BASIS OF SYMMETRY)

① ACTINOMORPHIC ⊕

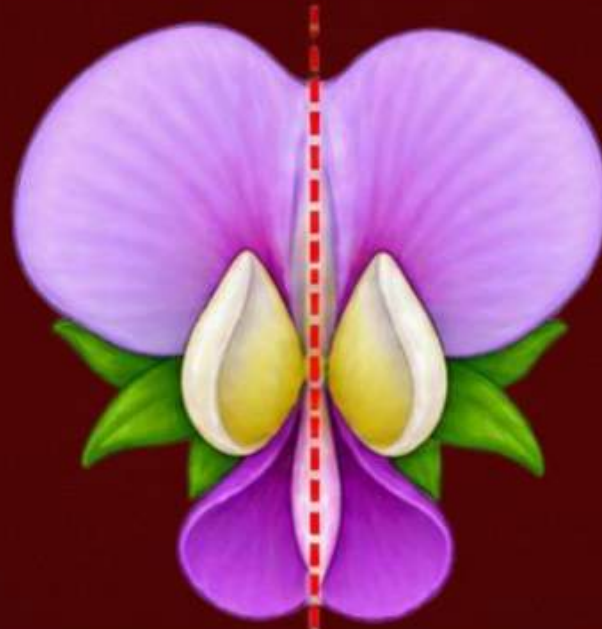
Mustard, Datura, Chilli,
Disc floret of sunflower



- Radial symmetry.
- Flower divided into two equal halves by any plane passing through centre.

② ZYGOMORPHIC %

Pea, Bean, Cassia, Gulmohar,
Ray florets of sunflower



- Bilateral symmetry.
- Flower divided into two equal halves by only one plane passing through centre.

③ ASYMMETRIC \$

Canna

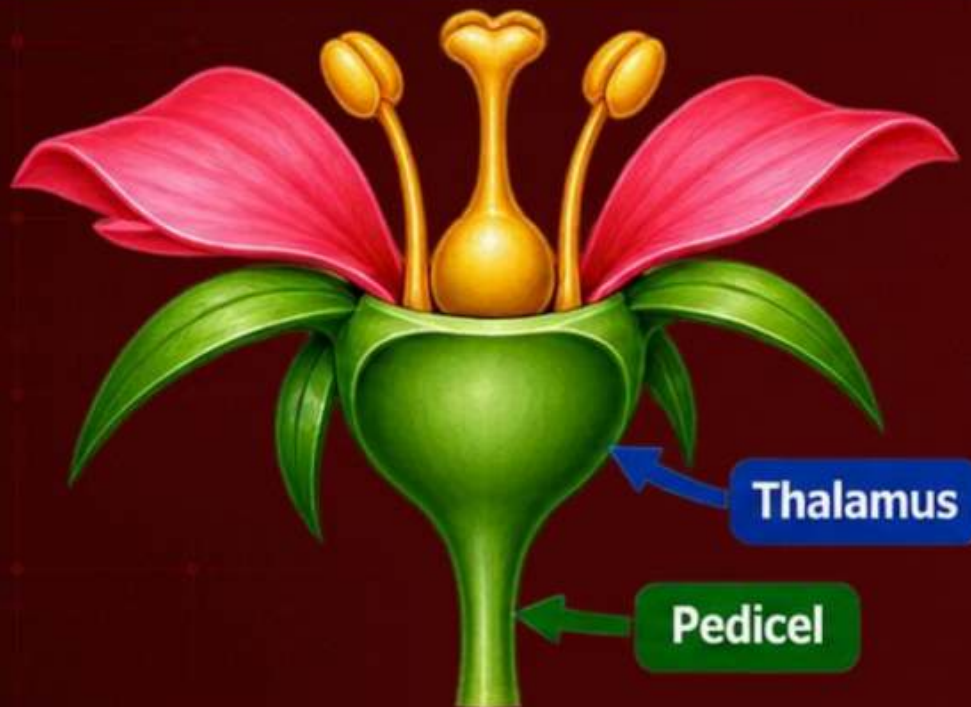


- No symmetry.
(Asymmetrical flowers)

FLOWER TYPES (ON THE BASIS OF OVARY POSITION)

① HYPOGYNOUS

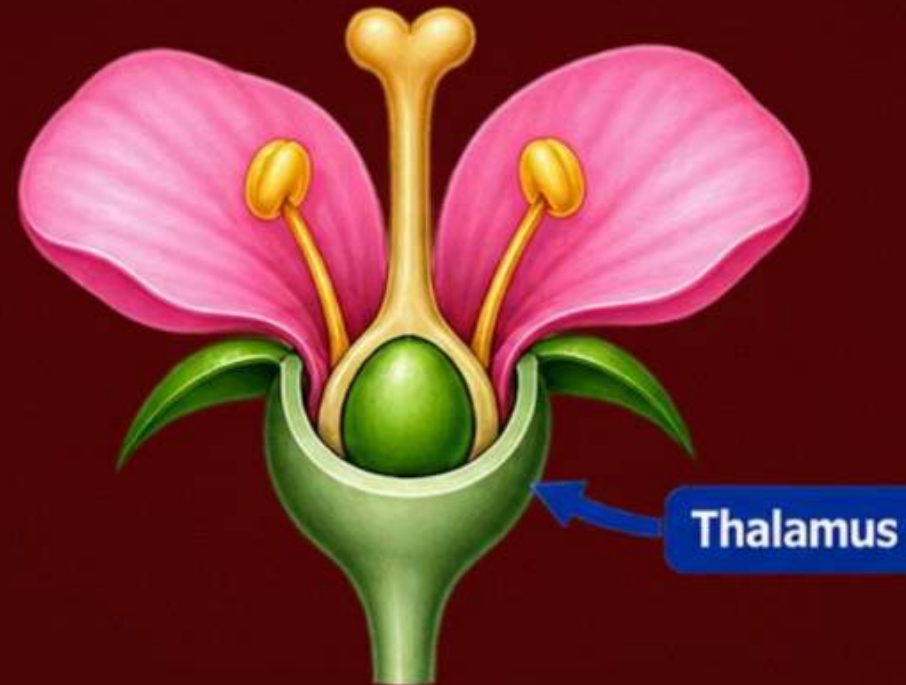
Mustard, China rose, Brinjal



- Ovary occupies top most position.
- Ovary is superior ($\underline{G}$).

② PERIGYNOUS

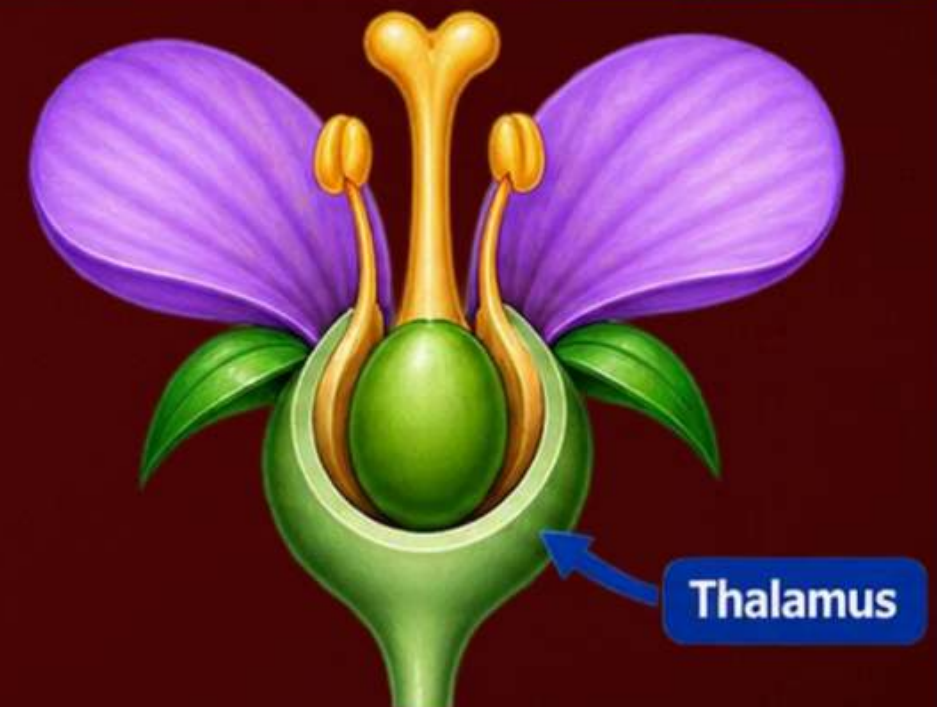
Plum, Rose, Peach



- Ovary partially inside thalamus.
- Ovary is half superior ($G -$).

③ EPIGYNOUS

Guava, Cucumber, Ray florets of sunflower



- Ovary completely inside thalamus.
- Ovary is inferior ($\overline{G}$).

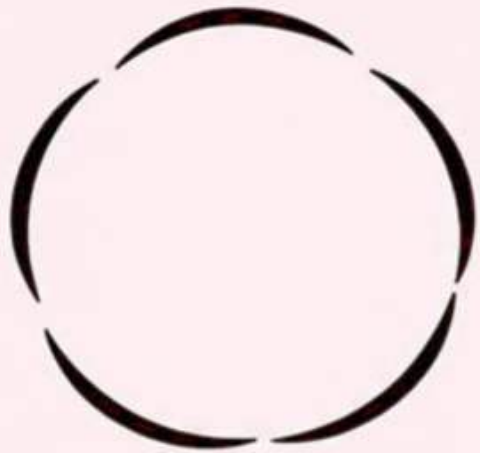
AESTIVATION

Pattern of arrangement of sepals/petals w.r.t. each other in a same floral whorl.

04 types :

① VALVATE

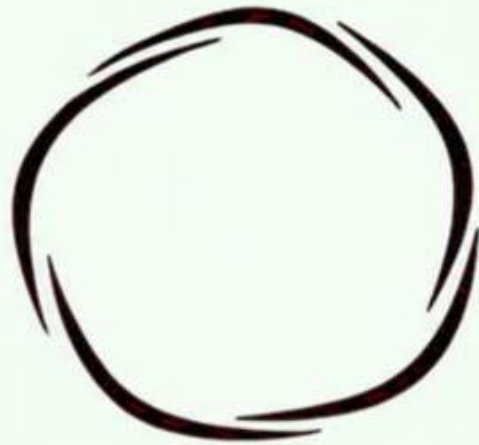
Calotropis



- Margins of sepals/petals just touch to each other (not overlap).

② TWISTED

China rose, Lady's finger, Cotton



- One margin toward outside of each sepal/petal.
- One margin toward inside.

③ IMBRICATE

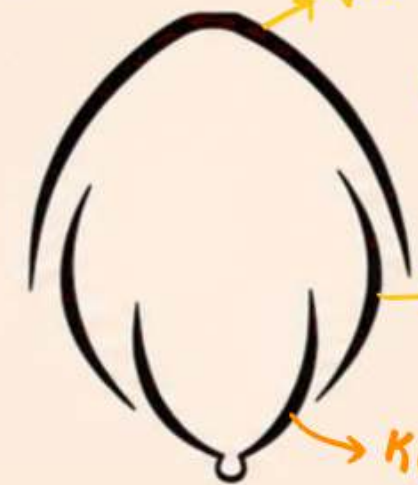
Cassia, Gulmohar



- One sepal/petal completely inside.
- One sepal/petal completely outside.
- Rest are like twisted aestivation.

④ VEXILLARY

Pea, Bean



- Total 5 petals (1 vexillum + 2 lateral + 2 fused keel).
- Lateral wings (2 in no.).
- Keel (2 in no.) fused (contion end).
- Seen only in petals.

Seen only in petals.

Vexillum/standard...
1 in no. (largest)

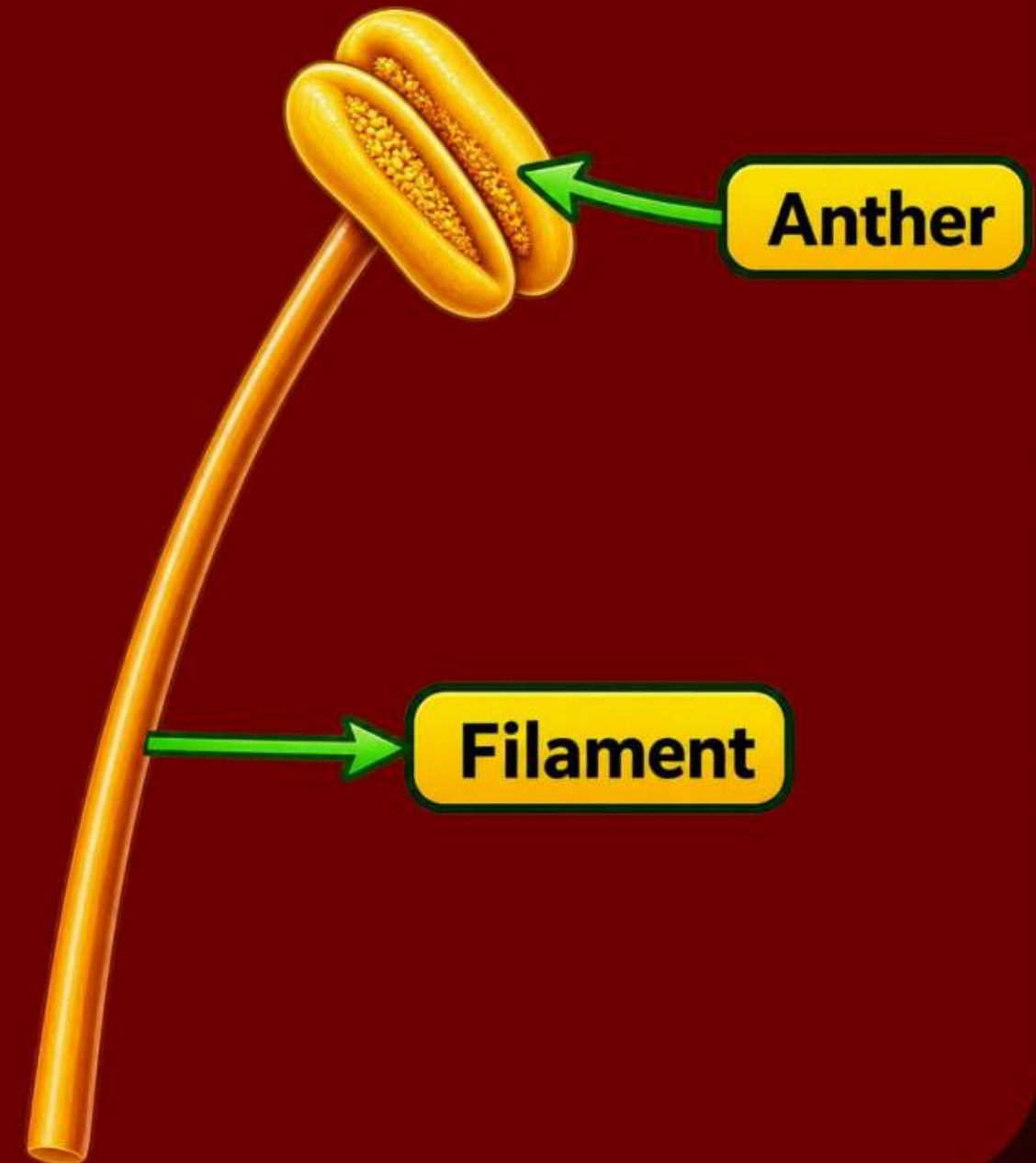
lateral wings
2 in no.

Keel (2 in no.)
fused.

Androecium (A)

- Male reproductive whorl.
- Group of stamens.
- Stamen — 2 parts:
- **Filament**
 - Stalk like structure
 - Sterile part.
- **Anther**
 - Fertile part
 - Produces pollen grain

Note: Sterile stamen — is called Staminode.



Adhesion of Stamen

Attachment of Stamen with other floral part is called as Adhesion of Stamen.

2 Types:

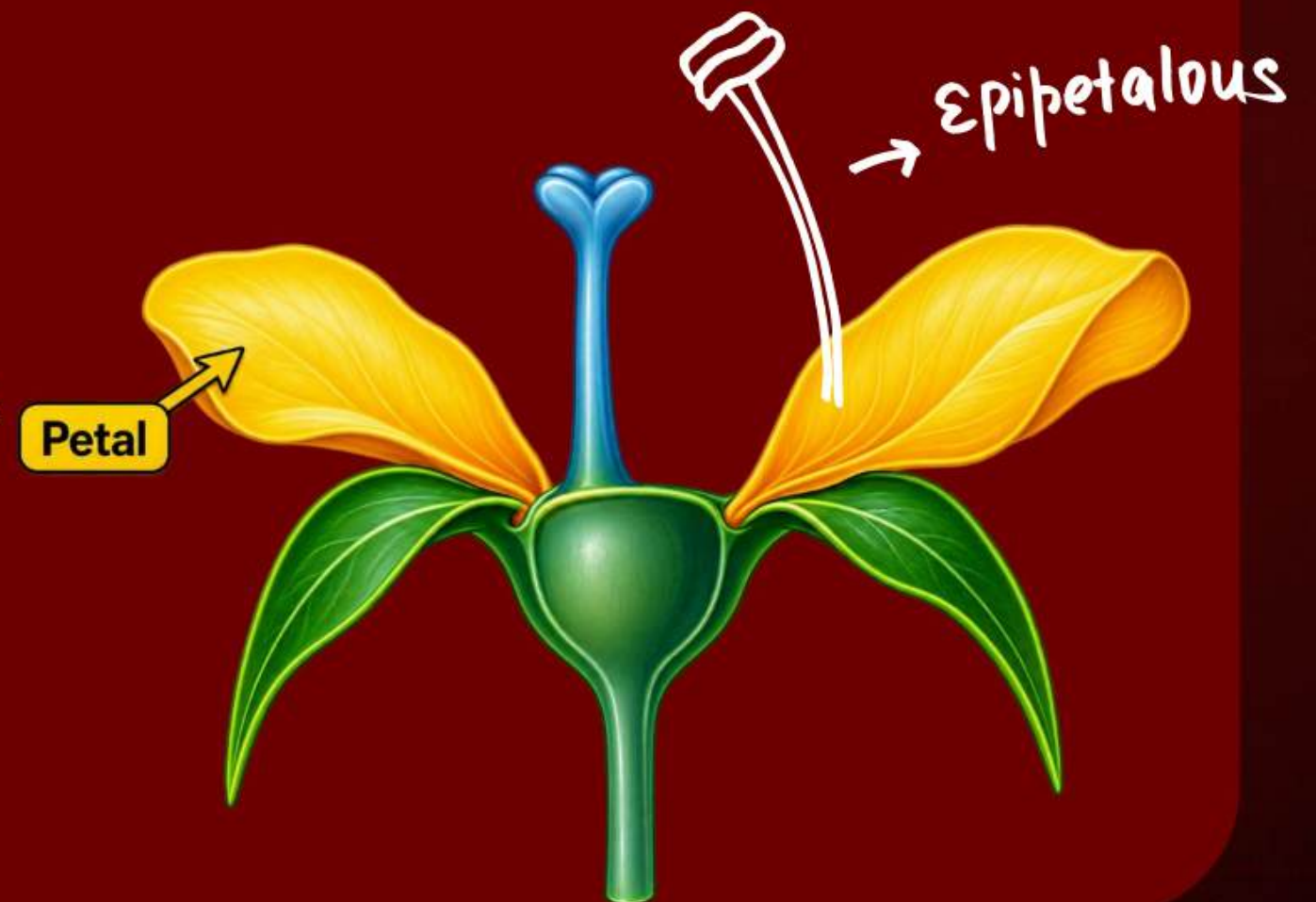
1. EPIPETALOUS:-

- Filament of Stamen is fused to petals.
- Represented by $\widehat{CA}$
- Ex. Brinjal (Solanaceae)

2. EPITEPALOUS:-

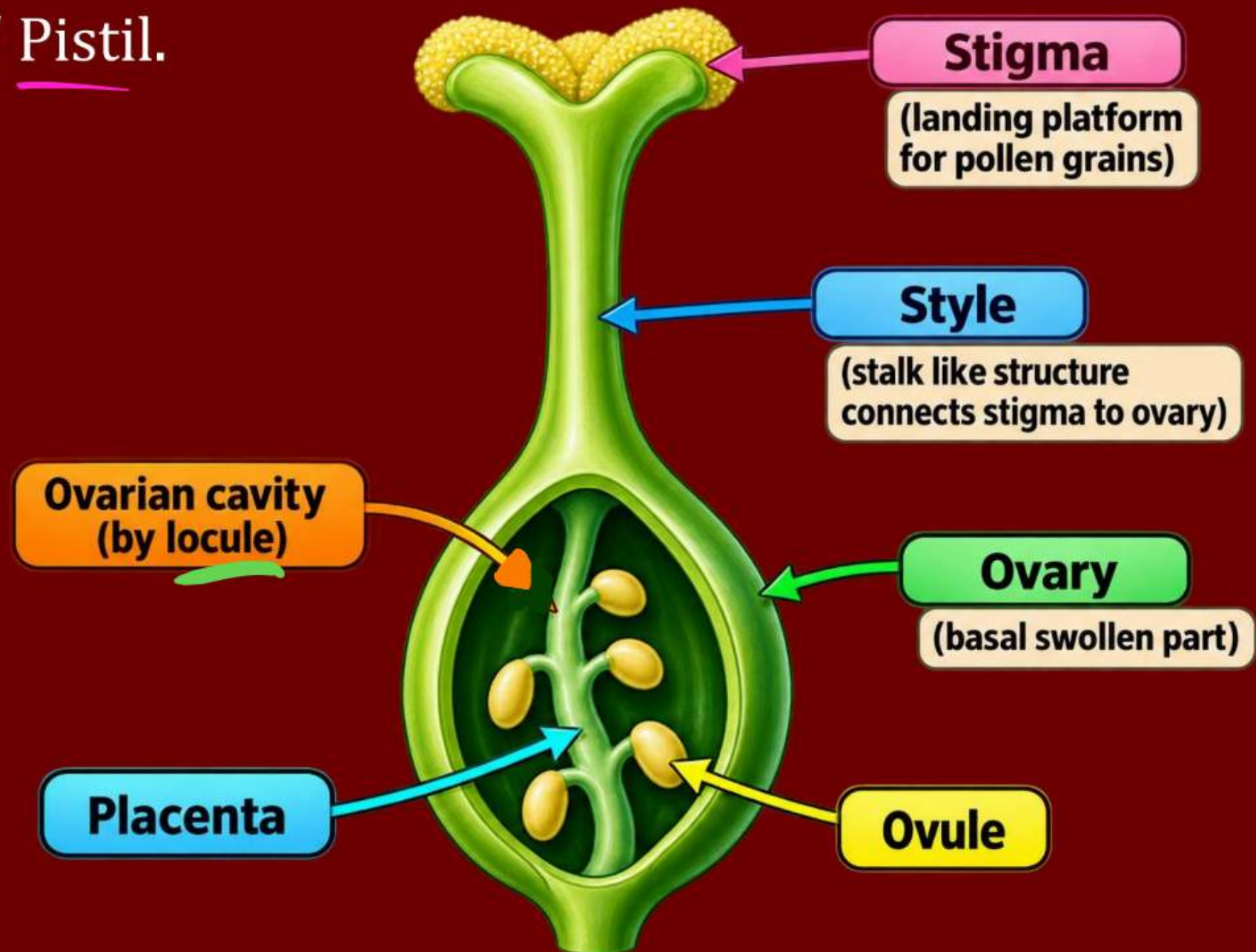
- Filaments of Stamen are fused to tepals.
- (Whorl of tepals — is kid Perianth)
- Represented by $\widehat{PA}$

Ex. Lilly.



Gynoecium

- Female reproductive whorl.
- Individual unit = Carpel / Pistil.
- Carpel — 03 parts



Gynoecium



- Single Carpel present in flower $\Rightarrow$ Monocarpellary.
- Multiple Carpels present in flower $\Rightarrow$ Multicarpellary.

Multicarpellary:

Fused $\Rightarrow$ Multicarpellary Syncarpous.

- Ex. Poppy, Mustard.

Free $\Rightarrow$ Multicarpellary Apocarpous.

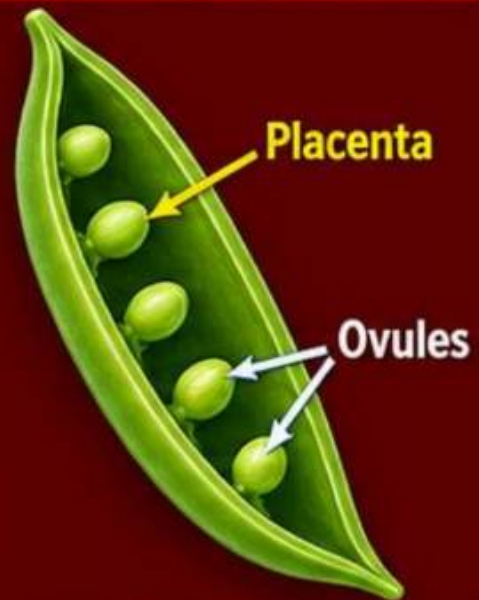
- Ex. Lotus, Rose

PLACENTATION

Pattern of arrangement of ovules on placenta inside ovary
05 types :

① MARGINAL

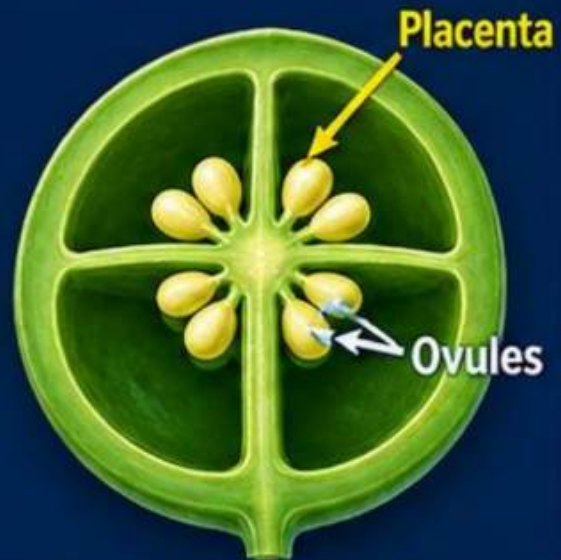
Pea



- Placenta bears ovules along ventral suture.
- Ovary unilocular.

② AXILE

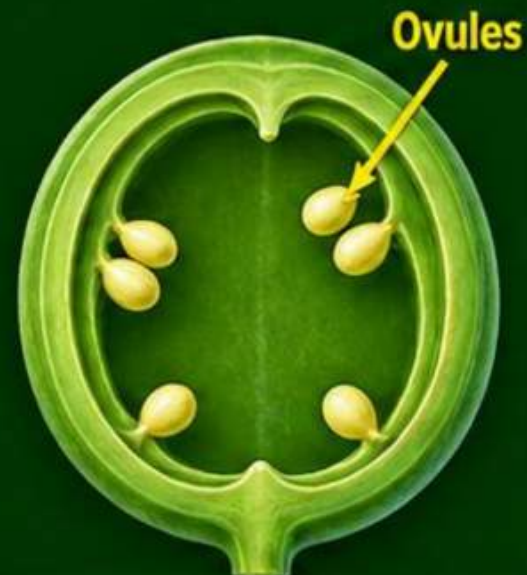
China rose,
Tomato, Lemon



- Placenta is on axial axis; ovules develop on it.
- Ovary multilocular due to presence of septa.

③ PARIETAL

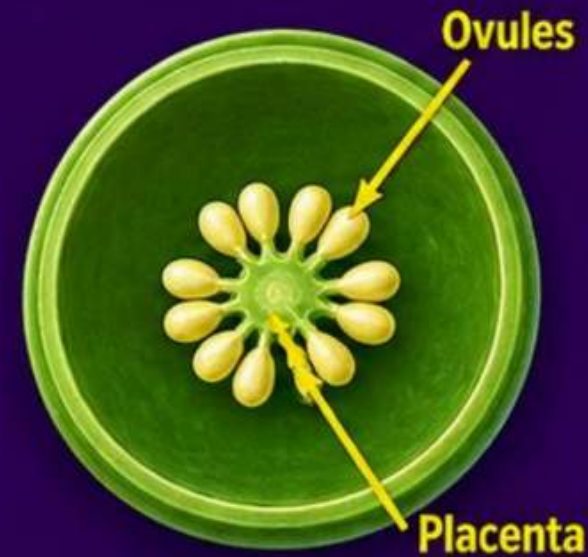
Mustard,
Argemone



- Placenta is peripheral; ovules develop at periphery.
- Ovary initially unilocular but later becomes multilocular due to false septum (replum).

④ FREE CENTRAL

Dianthus,
Primrose,



- Placenta is central in position and bears ovules.
- Ovary *unilocular*.

⑤ BASAL

Sunflower,
Marigold



- Most advanced.
- Single ovule develops at base of ovary.

Fruit



(k/d) → called

- Mature / ripened Ovary.
- Formed after fertilization.
- Fruit wall is k/d 'pericarp' develops from ovary wall.
- Pericarp — thick & fleshy, then it is differentiated into 3 layers:
 - a) Epicarp (outer) ✓
 - b) Mesocarp (middle) ✓
 - c) Endocarp (inner) ✓

Ex. Mango

- Mesocarp - juicy & fleshy.
- Endocarp - hard, leathery.
- Coconut — mesocarp is fibrous.
- Both mango & coconut develop from monocarpellary superior ovaries.

3 Types of Fruit

EUCARP

1 True Fruit

- Develops completely from ovary.
- Example:** Mango



pseudocarp

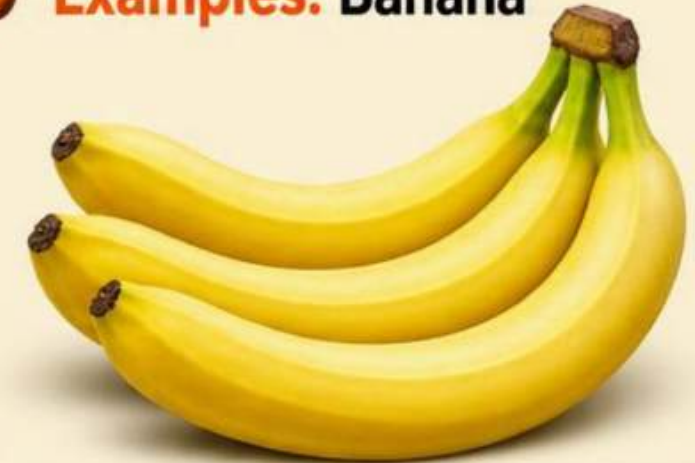
2 False Fruit

- Develops from other floral parts (like thalamus) along with ovary.
- Examples:** Apple, Strawberry, Cashew



3 Parthenocarpic Fruit

- Seedless fruit.
- Formed without fertilization.
- Examples:** Banana



Seed

SEED

- fertilized ovule.
- formed after fertilization.
- Two types:

a) Dicot Seed:

• Parts:

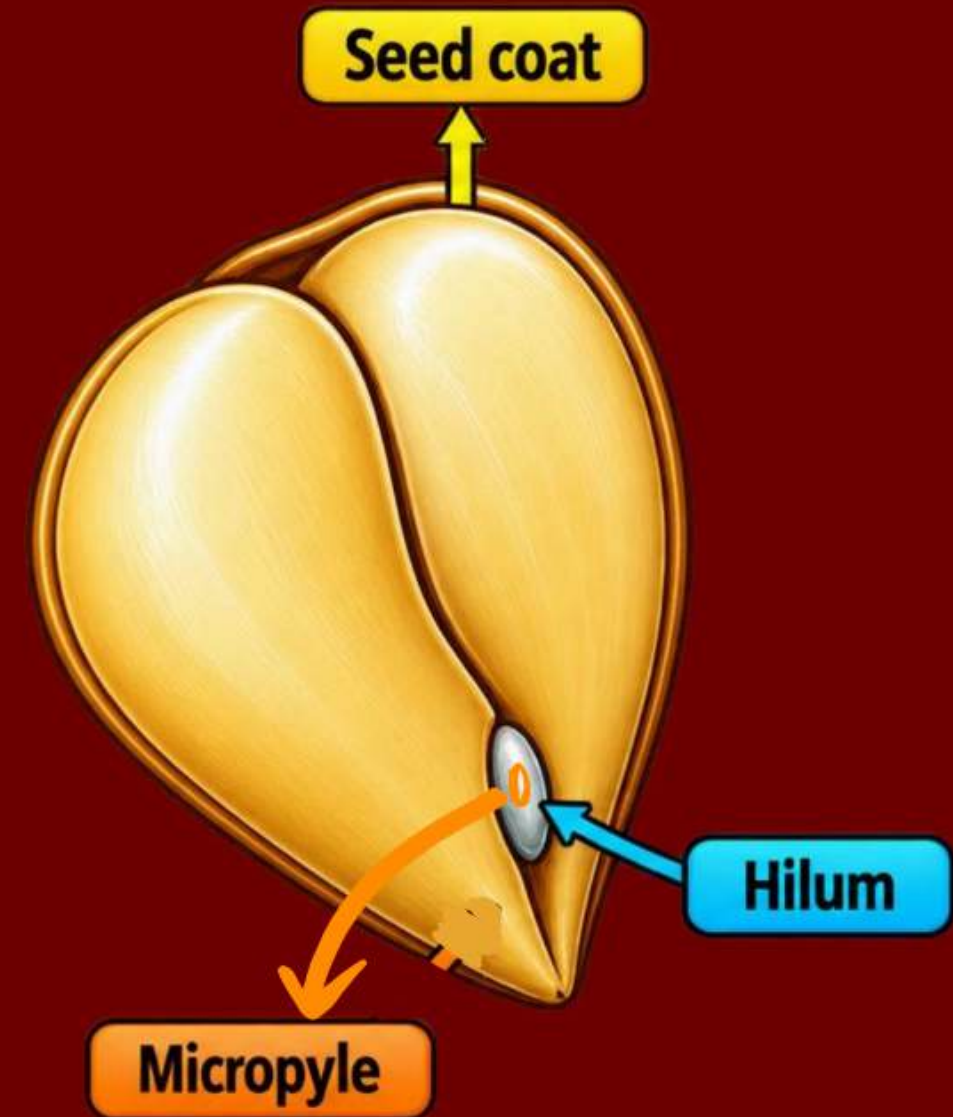
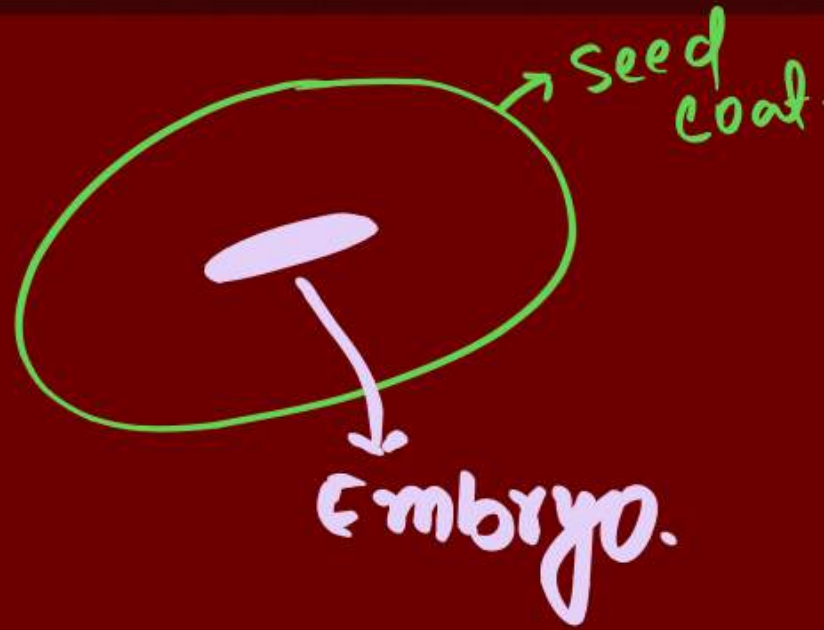
- a) Seed Coat: protective covering

2 layers:

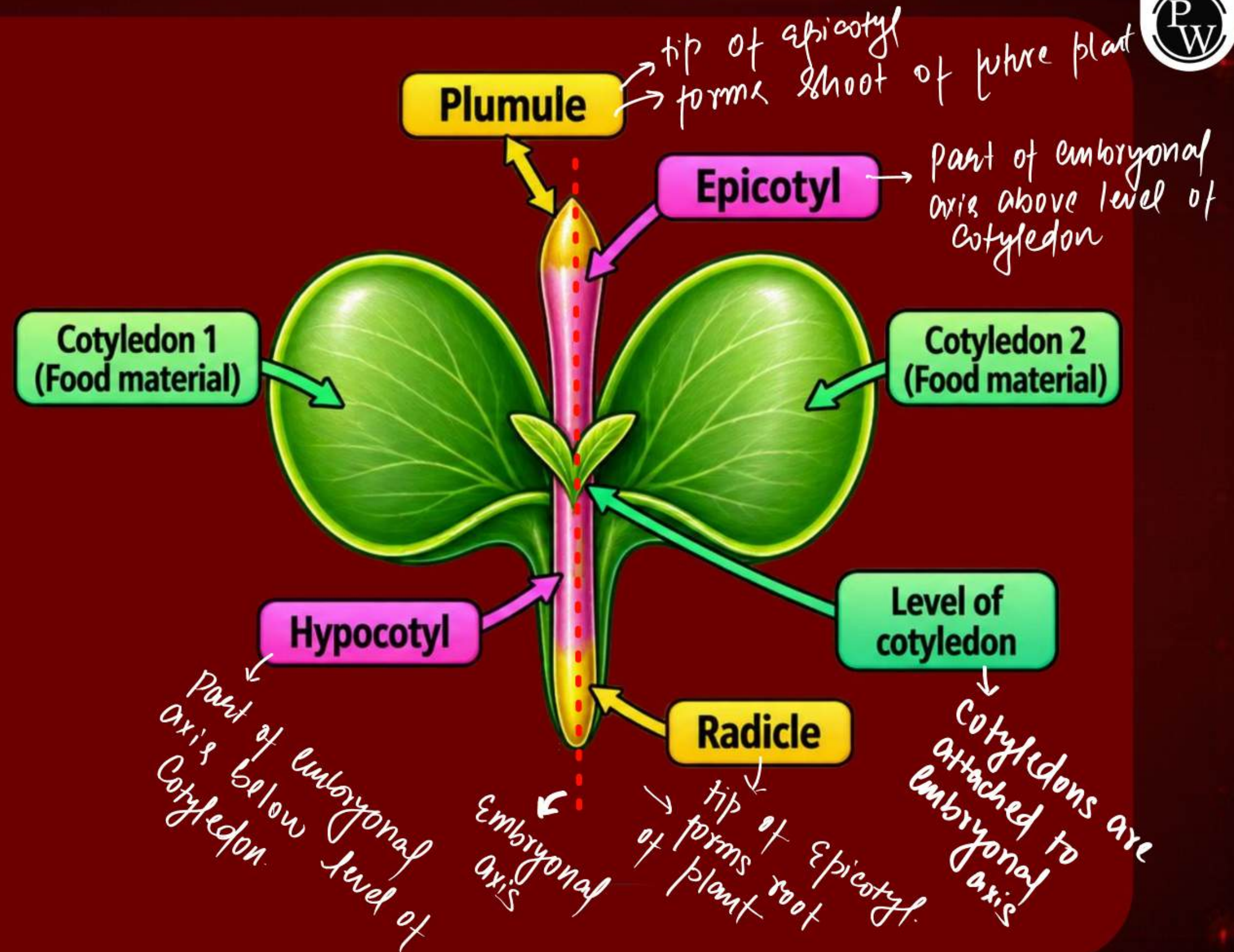
- outer - testa,
- inner - tegmen
- Poses scar like structure known as Hilum through which developing seed remain attached with fruit

Micropyle - Above hilum

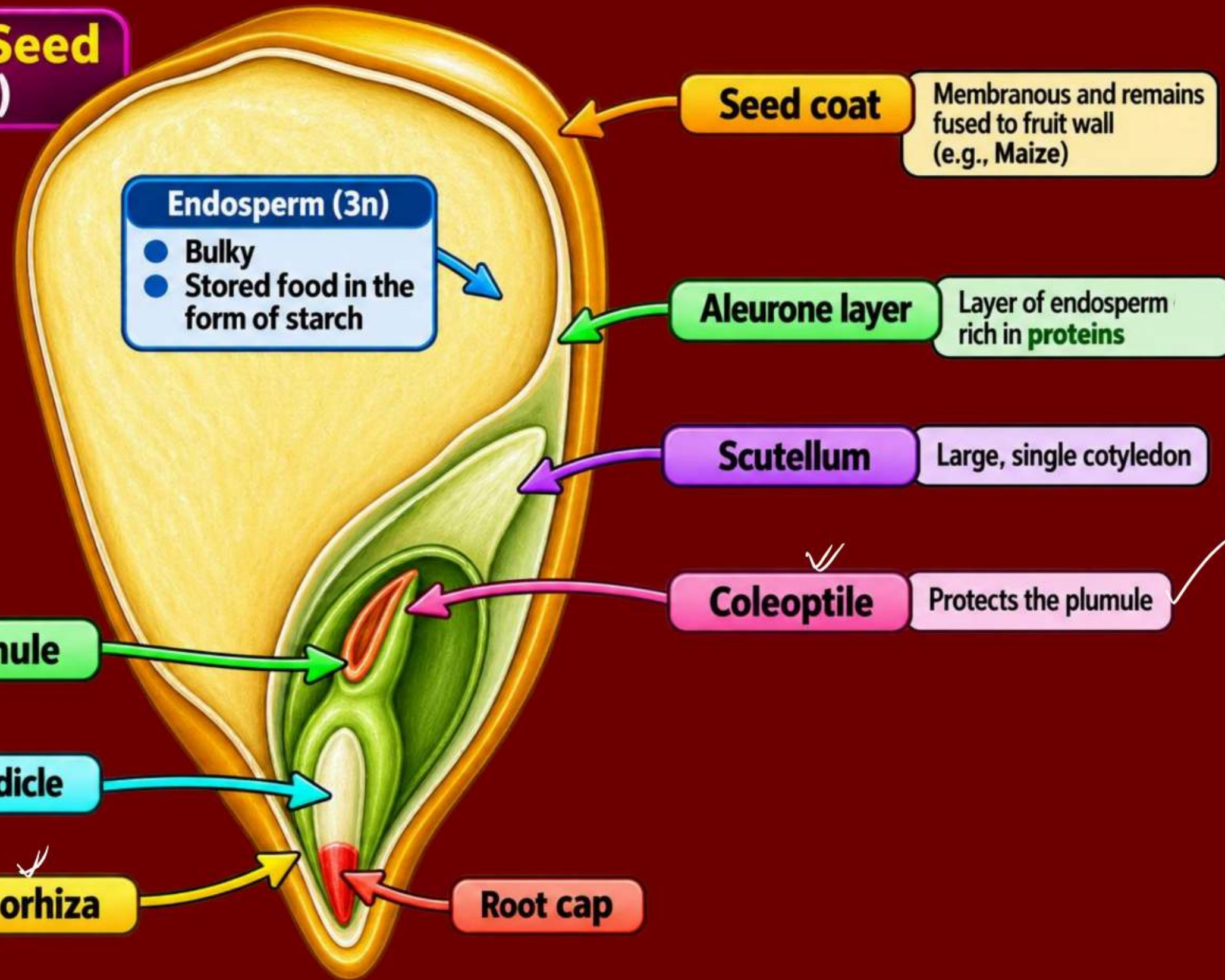
- pore for O_2 & H_2O entry during seed germination.



Embryo



Monocot Seed (Maize)



Family Solanaceae (Potato Family)

Vegetative Characters:

Stem: herbaceous, rarely woody, aerial, underground stem in potato (Solanum tuberosum)

Leaves: alternate, simple, rarely pinnately compound, exstipulate; venation reticulate.

↓
Stipule - nt

Family Solanaceae (Potato Family)

Floral Characters: ✓

Inflorescence: Solitary, axillary or cymose as in Solanum

Flower: bisexual, actinomorphic

Calyx: sepals five, united, persistent, valvate aestivation

Corolla: petals five, united; valvate aestivation

Androecium: stamens five, epipetalous

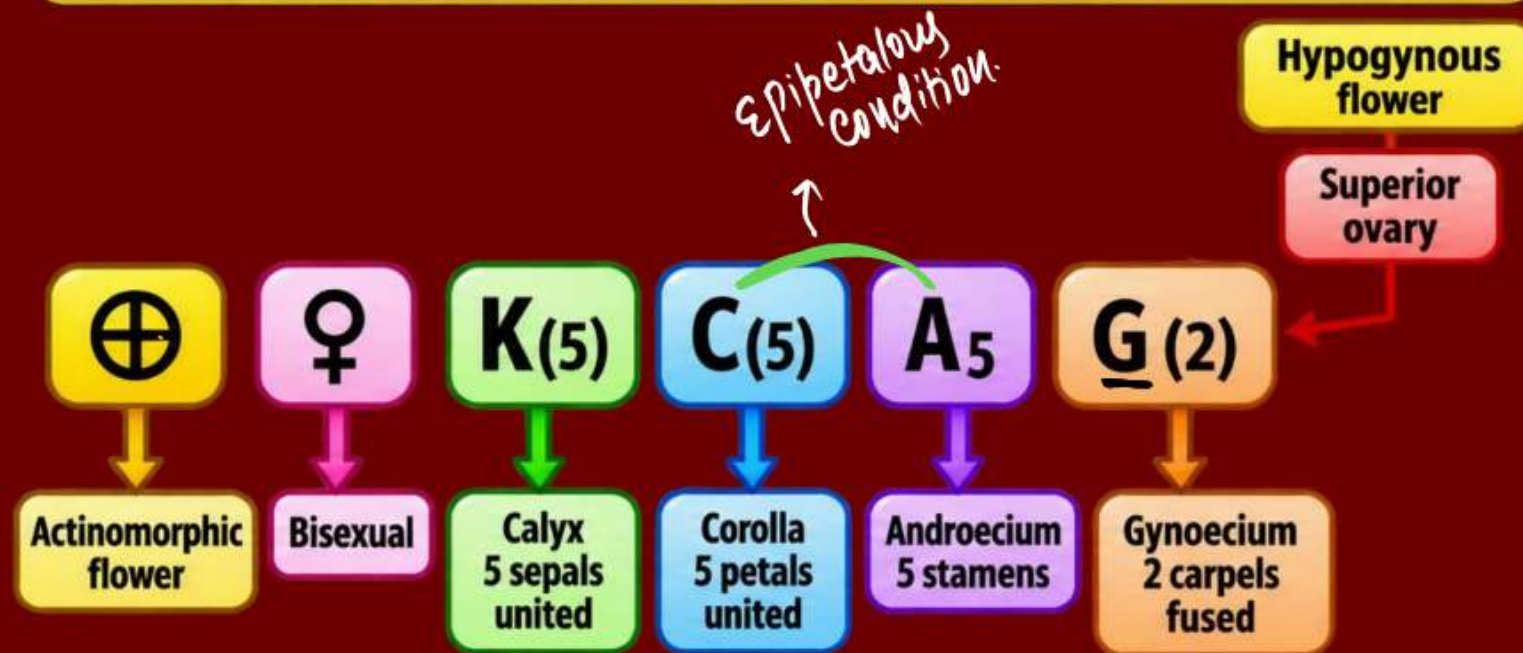
Gynoecium: bicarpellary obliquely placed, syncarpous; ovary superior, bilocular, axile placentation

Fruits: berry or capsule

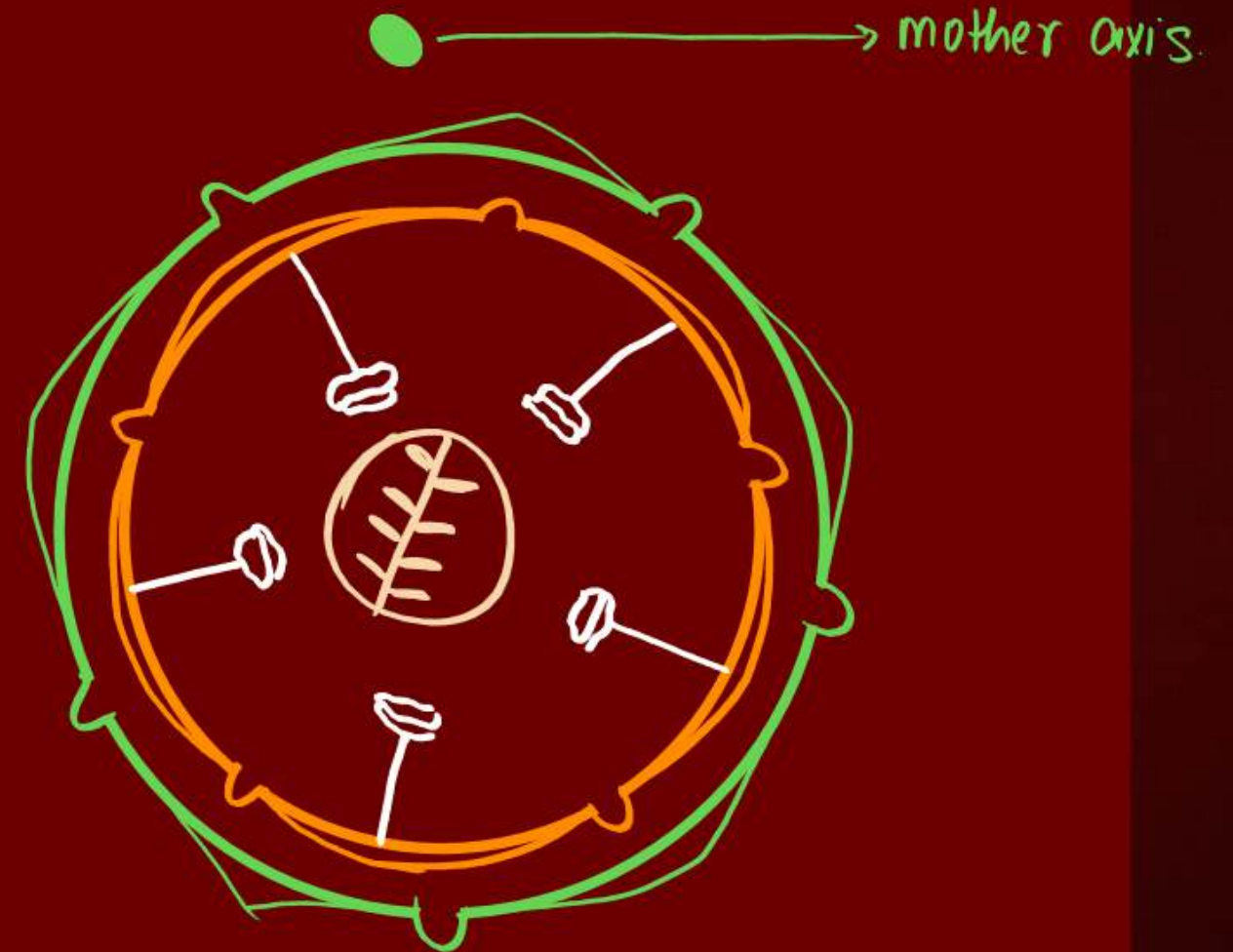
Seeds: many, endospermous.

Characteristic feature
of Solanaceae family.

FLORAL FORMULA



FLORAL DIAGRAM



Economic Importance

- Source of food e.g. tomato, brinjal, potato
- Spice e.g. chilli
- Medicine e.g. belladonna, ashwagandha
- Fumigatory e.g. tobacco
- Ornaments e.g. petunia.

Most IMPORTANT Questions for JKBOSE 2026

Q1. Discuss general characteristics of root & Regions of root?

Q2. Phyllotaxy & its types?

Q3. Inflorescence & its types?

Q4. What is flower? Write names of flower parts.

Q5. Monocot seed (maize seed) & Dicot seed?

Q6. Fruit.

Q7. Brief description of family Solanaceae?

HOMework



Revision



Thank
You